

ETHIOPIAN HEALTH PROFESSIONALS LICENSING EXAMINATION(EHPLE) PRACTICE TEST BOOKLET 2025

|ANESTHESIA



ጤና ሚኒስቴር - ኢትዮጵያ
MINISTRY OF HEALTH-ETHIOPIA

የዜጎች ጤና ለሃገር ብልጽግና!
HEALTHIER CITIZENS FOR PROSPEROUS NATION!



Message from the State Minister, Ministry of Health – Ethiopia



Mrs. Frehiwot Abebe

Improving the quality of healthcare remains a critical global priority and is essential for achieving sustainable development and universal health coverage. One of the most effective ways to ensure high standards in healthcare delivery is through the consistent assessment of professional competencies. In line with this commitment, the Ministry of Health of Ethiopia launched the Ethiopian Health Professionals Licensing Examination (EHPLE) in 2019 for undergraduates in seven core health disciplines. Since its inception, the program has expanded to encompass 17 health professions, reflecting our continued effort to build a competent and responsive health workforce.

The EHPLE was established with the primary aim of identifying and certifying health professionals who possess the minimum competencies required to provide safe and effective care. This ensures that all practicing professionals are equipped to meet the expectations of the communities they serve. The exam is coordinated by the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), which consists of four specialized desks. The LEO plays a vital role in system strengthening, enabling the exam to be conducted with increased efficiency, organization, and quality assurance.

It is important to recognize that the EHPLE is not a conventional academic or recruitment examination. Rather, it is a rigorous, competency-based assessment designed to evaluate the practical skills and critical thinking abilities that are essential for professional practice. This *Practice Items Booklet* is part of our broader strategy to support examinees and educators in preparing for the licensing process. It offers a realistic overview of the exam structure, promotes a deeper understanding of its objectives, and fosters transparency and sustainability in implementation.

The development of this booklet has been made possible through the collaborative efforts of experts from higher education institutions, the Addis Ababa University Testing Center, and professional associations. Their insights and dedication have been instrumental in shaping this resource. I extend my heartfelt thanks to all contributors and to the Ministry of Health team for their tireless work and steadfast commitment to excellence in healthcare regulation.

We believe this booklet will serve as a valuable tool for candidates preparing for the examination and for educators guiding them toward success. Together, we are building a stronger, more competent health workforce for Ethiopia.

Introduction EHPLE

Welcome to the 2025 Edition of the Ethiopian Health Professionals Licensing Examination Practice Test Booklet.

This resource is designed to support candidates in their preparation for the national licensing examination, ensuring they are well-equipped with the knowledge and clinical reasoning skills required to deliver competent and ethical healthcare across Ethiopia.

In Ethiopia, the licensing examination serves as a pivotal tool in evaluating the competency of health professionals who have completed their academic training and are ready to transition into independent practice. The exam not only ensures public safety by verifying professional readiness but also upholds national and international standards in healthcare service delivery. As the healthcare system continues to expand and evolve, a strong, well-regulated professional workforce becomes even more essential.

This practice booklet contains 200 multiple-choice questions modelled after the structure, content, and format of the actual licensing examination. The questions span across core clinical domains essential for entry-level practice. This includes areas such as Patient care, leadership and management, ethics, and Health Promotion as outlined in the Information Booklet 2024.

The booklet is intended to:

- Familiarize candidates with the structure and cognitive level of the actual licensing examination.
- Offer opportunities for self-assessment and identification of strengths and gaps in knowledge.
- Encourage critical thinking and clinical decision-making in scenarios reflective of real-world healthcare practice in Ethiopia.
- Reduce exam-related anxiety by providing realistic practice experiences.

Administered under the auspices of the Ministry of Health and relevant regulatory bodies, the Ethiopian Health Professionals Licensing Examination is grounded in evidence-based frameworks and aligned with both national needs and global best practices. It has been structured to assess a range of competencies including medical knowledge, communication, professionalism, and safe clinical practice.

This practice test is not a predictive tool, and performance on it does not guarantee similar outcomes on the actual exam. However, it provides an invaluable opportunity for familiarization with question styles, pacing strategies, and content review. We strongly recommend that candidates simulate real testing conditions while using this resource, including timing their responses and reviewing rationale for both correct and incorrect answers.

The goal of licensing examinations is not only to regulate entry into the profession but also to inspire excellence and lifelong learning. By engaging with this practice test, you are taking a proactive step toward professional development, accountability, and a commitment to serving your patients with competence and compassion.

We wish you success on your licensing journey—and beyond.

Acknowledgements

This *Practice Test Items Booklet for the Ethiopian Health Professionals Licensing Examination* is the result of a collaborative effort among dedicated professionals committed to advancing the quality of health sciences education in Ethiopia. The Ministry of Health extends its sincere appreciation to all those who contributed their time, expertise, and insights to the development of this essential resource.

Special recognition is given to the item developers and reviewers from various higher education institutions—both public and private—whose academic rigor and practical experience were instrumental in shaping the content and ensuring its relevance to real-world healthcare practice. Their contributions have helped create a high-quality, standardized practice tool that reflects national competencies and global benchmarks.

We are especially grateful to the team at the Addis Ababa University Testing Center, whose technical support and academic guidance were critical throughout the process. Their leadership in assessment design and educational evaluation has significantly strengthened the quality of this booklet.

We also thank the committed staff of the Ministry of Health, particularly those within the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (HHRIPR LEOs), whose coordination, oversight, and unwavering dedication ensured the successful completion of this project.

Together, these collective efforts reflect our shared vision of building a competent, ethical, and resilient health workforce for Ethiopia.

Anesthesia

1. A 28-year-old primigravida woman, who is in the 39th week of gestation, is presented to a labor ward with complaints of lower back pain and pushing down sensation of three hours duration. She has no antenatal follow-up. Physical examination reveals that her vital signs are within normal limits but there is a sign of uterine contraction. Per-vaginal (PV) examination shows a 5-centimeter dilated cervix. She prefers non-pharmacologic methods of labor pain relief.

What is the most preferable management plan for this woman?

- A. Hypnosis
- B. Aromatherapy
- C. Hydrotherapy
- D. Breathing techniques

ANSWER: C

2. A 55-year-old woman, who is diagnosed with stage II cervical cancer, is scheduled for a total abdominal hysterectomy under general anesthesia. She has received four cycles of unspecified chemotherapy. Her electrocardiogram result shows nonspecific ST-T changes and decreased QRS voltage, premature ventricular contractions, and supra-ventricular tachydysrhythmias. It was supposed to be an early toxicity of the drugs that might progress to cardiomyopathy.

What is the most likely medication causing this condition to be discontinued?

- A. Cisplatin
- B. Tamoxifen
- C. Doxorubicin
- D. Methotrexate

ANSWER: C

3. A 68-year-old female patient, who is diagnosed with endometrial cancer, is scheduled for a total abdominal hysterectomy under general anesthesia. She is a known type-II diabetic patient on oral hypoglycemic agents for the past four years.

What is the most appropriate indicator of optimum glycemic control for this patient?

- A. Hemoglobin A1C
- B. Fasting blood glucose level
- C. Post-prandial blood glucose level
- D. Random blood glucose level

ANSWER: A

4. A 25-year-old primigravida woman, who is at her 39th week of gestation and with a diagnosis of breach presentation, is scheduled for elective cesarean section under spinal anesthesia.

What is the most appropriate consideration during spinal anesthesia in this woman?

- A. Slower onset
- B. Increased duration
- C. Increased dose requirement
- D. Reduced dermatome spread

ANSWER: B

5. A 20-year-old primigravida woman, who is in her 39th week of gestation, is planned for emergency cesarean delivery. The pre-anesthetic evaluation revealed a history of bronchial asthma. Using physical examination, a blood pressure of 160/98 mmHg, pulse rate of 90/min, respiratory rate of 18/min, and temperature of 37 degrees Celsius was measured. Meanwhile, her blood pressure, heart rate and SpO₂ became 200/125mmHg, 63/min and 100 percent respectively. What is the most appropriate first-line drug in the management for this woman?

A. Hydralazine
B. Labetalol
C. Nifedipine
D. Esmolol

ANSWER: A

6. A known diabetic pregnant woman is booked for elective cesarean section. She is receiving a long-acting oral hypoglycemic agent. The anesthetist at the preadmission anesthesia clinic advised the parturient about the regimen change plan of the oral hypoglycemic agent. What is the most appropriate patient counseling plan for this case?

A. Continue till the morning day of surgery
B. Discontinue one day before surgery
C. Take half of the usual dose on the day of surgery
D. Discontinue starting from 5 days before surgery

ANSWER: B

7. A 34-year-old primigravida woman, who is at 38 weeks of gestation, is presented to an emergency department with complaints of severe headache, visual disturbance, and epigastric pain. On physical examination, her vital signs show a blood pressure of 170/110 mmHg, pulse rate of 90/min, respiratory rate of 22/min, and temperature of 37 degrees Celsius. She had abnormal body movement and a brief loss of consciousness 30 minutes ago. Laboratory results showed hemoglobin of 9 g/dl, hematocrit of 27 percent, and platelet count of 96,000 x 10 the power of 9/L. In addition, coagulation studies were requested to be obtained before commencing anesthesia.

What is the most likely concern to look at additional laboratory results for this case?

A. Thrombocytopenia
B. Possibility of severe pulmonary edema
C. Disseminated intravascular coagulopathy
D. To determine the need for invasive central hemodynamic monitoring

ANSWER: C

8. A 38-year-old multiparous woman with two previous cesarean deliveries is scheduled for elective cesarean delivery. The vital signs and laboratory results are within the normal limits and there is no coexisting medical illness. The parturient prefers general anesthesia complaining that the previous anesthesia was painful due to multiple attempts.

What is the most likely anatomic change contributing to the difficulty of lumbar puncture in this case?

A. Widening of the pelvis
B. Lumbar lordosis
C. Thoracic kyphosis
D. Widened interspinous gap

ANSWER: B

9. A 25-year-old woman, with active labor at 40 weeks of gestational age, is diagnosed with fetal bradycardia and decided for emergency cesarean delivery. On admission to an operation room, her vital signs showed a blood pressure of 90/60mmhg, heart rate of 90/min, respiratory rate of 16/min, SpO₂ of 98 percent, and fetal heartbeat of 70/min. Laboratory investigation findings showed a hematocrit level of 38 percent, a WBC count of 10,000/mm³, a platelet count of 70 × 10⁹/L.

What is the most appropriate anesthetic technique plan for this case?

- A. General anesthesia with RSI
- B. Combined spinal-epidural anesthesia
- C. Conventional general anesthesia
- D. Rapid Sequence spinal anesthesia

ANSWER: A

10. A 21-year-old term parturient, with the diagnosis of fetal bradycardia, is planned to undergo cesarean delivery. She complained difficulty of breathing in a supine position and voice changes. Physical examination shows a short neck, a sternomental distance of 10cm, mouth opening of two fingers, jaw slide grade-B, and oropharyngeal view grade-II. Her blood pressure is 90/55mmHg, heart rate is 80/min, SpO₂ is 92 percent, and fetal heart rate is 70/min.

What is the most appropriate airway management plan for this case?

- A. Intubate following standard induction of anesthesia
- B. Maintain spontaneous breathing until the airway is secured
- C. Proceed with awake tracheostomy
- D. Avoid intubation and administer spinal anesthesia

ANSWER: B

11. A 25-year-old Gravida 1 Para 0 parturient, at 38 weeks of gestation, presented to a hospital to undergo elective cesarean section. She has no previous surgical and anesthesia exposure. On physical examination, she is healthy-looking. A blood pressure of 110/70mmhg, heart rate of 90/min, and respiratory rate of 16/min was noted. After preoperative evaluation, the anesthetist took consent for spinal anesthesia with lidocaine, looking for an adjuvant which increases its alkalization (PH) to enhance the onset.

What is the most appropriate adjuvant drug to be prepared in this case?

- A. Sodium bicarbonate
- B. Neostigmine
- C. Clonidine
- D. Epinephrine

ANSWER: A

12. A 34-year-old parturient at 42 weeks of gestation came to a hospital to undergo elective cesarean delivery for indication of post-term with oligohydramnios. On history taking, she reported that she has a previous history of cesarean delivery scar. Physical examination findings show a blood pressure of 100/70mmhg, heart rate of 70/min, and SpO₂ of 92 percent. The parturient requested an epidural anesthesia and consent was taken,

What is the most appropriate initial step to be applied in compliance with patient safety?

- A. Patient-procedure verification
- B. Patient positioning
- C. Strict aseptic technique
- D. Needle and catheter placement

ANSWER: A

13. A 22-year-old parturient at 39 weeks of gestation age planned for elective cesarean delivery with a previous scar and opted to repeat. Physical examination findings show a blood pressure of 100/70mmhg, heart rate of 75/min, SpO₂ of 94 percent, a weight of 70kg, and height of 1.65m. Under strict aseptic technique, fluid preload and spinal anesthesia were given. What is the most appropriate initial sensory blockage assessment method in this case?
- Ask the patient to raise her leg
 - By looking at the drop in blood pressure
 - Apply a drop of cold saline or alcohol
 - Pin prick using a sharp needle
- ANSWER: C
14. A 20-year-old primigravida, who is diagnosed with prolonged labor, came for emergency cesarean delivery. Physical examination findings show a blood pressure of 100/70mmhg, heart rate of 75/min, and SpO₂ of 97 percent. Laboratory results are all in the normal range. After preloaded with 800ml of normal saline, spinal anesthesia with hyperbaric bupivacaine is administered following a strict aseptic technique. Fifteen minutes following the start of surgery, her blood pressure, heart rate, and SpO₂ became 80/60mmhg, 65/min, and 97 percent respectively. What is the most appropriate initial management?
- Head down and support with oxygen
 - Administer ephedrine
 - Give fluid with a fast infusion
 - Administer Atropine
- ANSWER: C
15. A 30-year-old parturient at her 38th week of gestation was referred to a hospital with a diagnosis of fetal distress to undergo cesarean delivery under general anesthesia. Her vital signs show blood pressure of 100/70mmhg, heart rate of 75/min, SpO₂ of 94 percent and fetal heartbeat of 100/min. Laboratory investigation findings are all normal. Five minutes later, she gave birth but the baby was not crying, gasping to breathe, and floppy with an Apgar score of 4. The baby produces weak sounds while dried, stimulated, and wrapped with no improvement. Then, the baby was taken to radiant warmer up. On further examination, the heart rate is found to be 80/min. What is the immediate next management step for this baby?
- Open intravenous access
 - Repeat stimulation until baby cries
 - Start chest compression
 - Give bag-mask ventilation support
- ANSWER: D
16. A 40-year-old mother came to a hospital with a chief complaint of protruding vaginal content and was later diagnosed with utero vaginal prolapse. Then she was scheduled for vaginal hysterectomy under spinal anesthesia. Accordingly, baseline vital sign was taken and spinal anesthesia with bupivacaine 15mg was administered between L3 and L4. The patient was positioned at lithotomy position for two hours with the lowest vital sign reading; blood pressure of 100/70mmhg, heart rate of 75/min, and SpO₂ of 94 percent. Eight hours into the postoperative period, she complained of a tingling sensation at the posterior part of the lower leg. What is the most likely injured nerve in this case?
- Posterior cutaneous nerve
 - Lateral cutaneous nerve
 - Common peroneal nerve
 - Saphenous nerve
- ANSWER: C

17. A 35-year-old parturient, who is a known asthmatic patient, was referred from a health center at her 36th week of gestation with a diagnosis of severe preeclampsia for emergency cesarean section. The right upper quadrant pain is noticed during physical examination. Baseline vital signs revealed a blood pressure of 160/100mmHg, heart rate of 80/min, and SpO₂ of 92 percent. Laboratory examination findings revealed a hematocrit level of 38 percent, platelet count of $70 \times 10^9/L$, and INR of 1.8. The parturient is on magnesium infusion started in the labor ward. The anesthetist is wondering about the drug interaction to select muscle relaxant.

What is the most appropriate dosage to be administered for muscle relaxation maintenance?

- A. Low dose of short-acting
- B. Routine dose of any type
- C. Low dose of long-acting
- D. High dose of any type

ANSWER: A

18. A 30-year-old term parturient was admitted to a labor ward after spontaneous vaginal delivery. Six hours later, the mother had profuse vaginal bleeding, later diagnosed as primary postpartum hemorrhage. The surgeon rushed to perform surgery with the possibility of an emergency hysterectomy. Her vital signs are; blood pressure of 70/50 mmhg, heart rate of 80/min, and SpO₂ of 96 percent. Laboratory investigation findings show hematocrit level of 18 percent, platelet count of $120 \times 10^9/L$, and INR of 1.2.

What is the most appropriate initial management plan for this case?

- A. Induce with a sufficient dose of ketamine and intubate
- B. Administer one bag of normal saline before anesthesia
- C. Request for lab determination of hemoglobin concentration
- D. Preoperative assessment should be bypassed

ANSWER: B

19. A 30-year-old term-sized gravid uterus mother with active labor was referred to a hospital. She reported that she has dyspnea on exertion and chest pain. Her vital signs show blood pressure of 90/60mmHg, heart rate of 90/min, SpO₂ is 90 percent and a loud systolic murmur was heard at the right upper sternal border which radiates to the carotid artery. On investigation, cardiac catheterization revealed an Aortic valve area of 2.5 cm² (normal 3.0 – 4.0) and a pressure gradient across the valve of 30mmHg (normal less than 25mmHg).

What is the most appropriate intraoperative anesthetic technique in this case?

- A. Spinal anesthesia with dense blockage
- B. Epidural anesthesia with preloaded fluid
- C. General anesthesia with RSI
- D. local infiltration with TAP block

ANSWER: B

20. A 25-year-old woman diagnosed with invasive molar pregnancy of 13 weeks of gestation was admitted to a hospital and later scheduled for suction curettage. She has a history of hot intolerance. Her vital signs show blood pressure of 100/70mmhg, heart rate of 95/min, and SpO₂ of 94 percent. Laboratory investigation shows a hematocrit level of 38 percent, while Thyroid function tests T3, T4, and TSH are in the normal range.

What is the most appropriate intraoperative anesthesia management plan for this case?

- A. Achieve a deep level of anesthesia with inhalational agent MAC greater than one percent
- B. Spinal anesthesia with sensory block level of T6 (mid thoracic) is needed
- C. There is a high possibility of trophoblastic mater embolization
- D. Use general anesthesia with low MAC less than 1 percent inhalational agent

ANSWER: D

21. A 31-year-old mother with a 24-week gestation age came with a diagnosis of acute appendicitis and planned for appendectomy. Examination findings show blood pressure of 110/70mmhg, heart rate of 95/min, and SpO₂ of 94 percent. Laboratory investigation results are all in the normal range.

What is the safest anesthetic technique plan in this case?

- A. Spinal anesthesia
- B. General anesthesia with RSI
- C. Epidural anesthesia
- D. Combined epidural-spinal anesthesia

ANSWER: A

22. A 27-year-old parturient at 14 weeks of gestational age sustained a mid-clavicle fracture. The patient is brought to an operation room for open reduction and internal fixation. Her vital sign shows blood pressure of 120/75mmhg, heart rate of 75/min, and SpO₂ of 97 percent. Laboratory investigation finding shows a hematocrit level of 39 percent. The patient is anxious and general anesthesia was planned with premedication.

What is the most appropriate premedication drug for this patient?

- A. Diclofenac
- B. Diazepam
- C. Morphine
- D. Midazolam

ANSWER: C

23. A healthy pregnant woman, at term, underwent a cesarean section under spinal anesthesia using 12mg bupivacaine. The surgery was uneventful. After staying in the recovery room, she was transferred to the maternity ward. On the third post-operative day, she started to complain of positional headache in the frontal region, diplopia, neck stiffness, nausea, and tinnitus. The pain was not relieved with the initial management approach.

What is the most reliable method to relieve these symptoms?

- A. Epidural morphine
- B. Intravenous caffeine
- C. Epidural blood patch
- D. Intravenous hydrocortisone

ANSWER: C

24. A 35-year-old primigravida, with a BMI of 30, is brought to an operation room to undergo an emergency cesarean section due to fetal distress. She has no pertinent medical history. Preanesthetic evaluation revealed vital signs are within the normal range and clear chest. Rapid sequence induction was done and endotracheal intubation was successful at the fourth attempt. There was a witnessed aspiration of gastric content during the attempt and SapO₂ became 85 percent, wheezing was heard during auscultation.

What is the most appropriate initial management plan for this case?

- A. Suctioning the airway
- B. Lung lavage with saline
- C. Prophylactic antibiotics
- D. Hydrocortisone

ANSWER: A

25. A 33-year-old woman, with a BMI of 30kg/m^2 , arrived at a post-anesthesia care unit (PACU) after undergoing an uneventful emergency cesarean delivery under general anesthesia. At PACU, monitors showed blood pressure of 145/95mmHg, heart rate of 120/min, and SaPo₂ of 89 percent. What is the most appropriate position to improve saturation for this case?
- A. Left lateral
 - B. Right lateral
 - C. Semi-Fowler's
 - D. Trendelenburg
- ANSWER: C
26. A 31-year-old, 60 kg, woman is transferred to a post-anesthesia care unit after undergoing cesarean delivery under general anesthesia. She has received propofol 120 mg, morphine 8 mg a, Suxamethonium 120mg, and halothane 0.5 vol percent. On arrival, her vital signs were within the normal range. A few minutes later, her blood pressure became 145/95 mmHg, pulse rate 117/min, SaPO₂ 87 percent, and respiratory rate 8/min with adequate tidal volume. She is somnolent but responsive to verbal and physical stimuli. What is the most appropriate drug to manage this case?
- A. Neostigmine
 - B. Flumazepin
 - C. Naloxone
 - D. Labetalol
- ANSWER: C
27. A 31-year-old primigravida woman is brought to an operation room to undergo an emergency cesarean section due to placenta abruption. On arrival, she was drowsy, blood pressure was 160/100 mmHg, pulse rate was 125/min, respiratory rate was 24/min, and SaPO₂ 89 percent. During auscultation, there was crackle and rale sounds bilaterally. There was significant blood loss during surgery. Three units of blood was transfused and intravenous vasopressor support started. Despite the conservative management, it became difficult to stop the bleeding, and then a peripartum hysterectomy was done. What is the most appropriate postoperative care plan for this woman?
- A. Put on vasopressor infusion and conduct deep extubation
 - B. Transfer to post-anesthesia care unit after extubation
 - C. Transfer to intensive care unit with an endotracheal tube
 - D. Keep her in the operation room until her hemodynamics are stable
- ANSWER: C
28. A 29-year-old woman is transferred to post-anesthesia care unit after undergoing cesarean delivery under spinal anesthesia and being infused with oxytocin 20IU in 500ml of Normal Saline. She has no pertinent past medical history. On arrival, her blood pressure was 100/60mmHg, pulse rate was 100/min, and respiratory rate was 16/min. A few minutes later, her blood pressure became 60/40mmHg, her pulse rate became 120/min and there was excessive bleeding through the vagina. What is the most preferable vasopressor for this case?
- A. Ephedrine
 - B. Epinephrine
 - C. Phenylephrine
 - D. Norepinephrine
- ANSWER: C

29. A 70-kg male patient, who is diagnosed with right upper lobe lung fibrosis, is scheduled for upper lobe lung resection. The spirometer result shows a forced expiratory volume in 1 second (FEV1) of less than 800 mL. A ratio of residual volume to total lung capacity is greater than 50 percent, maximal oxygen consumption (VO₂ max) is greater than 10 mL/kg/min, and tidal volume is 80ml/minute. Finally, the surgical team postponed the planned procedure. What is the most likely spirometer result that will support the clinical decision of the surgical team?
- A. Tidal Volume
 - B. Maximal oxygen consumption
 - C. Forced expiratory volume in 1 second
 - D. A ratio of residual volume to total lung capacity
- ANSWER: C
30. A 60-year-old known COPD male patient was scheduled for a left lung lobectomy. During the preoperative assessment, the patient reported difficulty of breathing, chest pain, and cough of three months duration. The anesthesia team decides to postpone the surgery to optimize the patient's condition and reduce postoperative respiratory complications. What is the most appropriate optimization strategy for this patient management?
- A. Administration of corticosteroids
 - B. Provision of chest physiotherapy
 - C. Administration of antibiotics
 - D. Administration of anticholinergic bronchodilators
- ANSWER: B
31. A 55-year-old male smoker and known hypertensive patient was brought to an outpatient department with complaints of shortness of breath and dyspnea on exertion. He was diagnosed with lung cancer, and scheduled for resection. The patient had undergone surgery for lung cancer on the contralateral lung. The anesthetist planned to intubate using a single-lumen tube with a bronchial blocker. What is the most likely type of bronchial blocker used for this patient?
- A. EZ Bronchial Blocker
 - B. Cohen Bronchial Blocker
 - C. Fuji endobronchial Uniblocker
 - D. Arndt Bronchial Blocker
- ANSWER: D
32. A 30-year-old female patient presented to an emergency outpatient department after sustaining a bullet injury to the right side of her chest two hours ago. Preoperative evaluation was done, and she underwent emergency thoracotomy with general anesthesia and a double-lumen tube in the lateral position. What is the most likely physiologic change this patient will face in the intraoperative time?
- A. The right lung is more ventilated and more perfused
 - B. The right lung is more perfused and less ventilated
 - C. The left lung is more ventilated and more perfused
 - D. The left lung is less ventilated and more perfused
- ANSWER: D

33. A six-year-old male child presented to an emergency outpatient department with a complaint of foreign body aspiration in the throat 10 days ago. During evaluation, the patient has a cough, hemoptysis, and occasionally experiences shortness of breath. Radiographic findings show the aspirated material at the lower border of the trachea. The surgical team planned to remove the foreign body, and the estimated duration of surgery is two hours.
What is the most appropriate type of anesthesia to be used in this case?
- A. General anesthesia using LMA
 - B. General anesthesia with endotracheal intubation
 - C. Midazolam sedation under spontaneous ventilation
 - D. Superficial cervical nerve blockers with supplemental sedation
- ANSWER: B
34. A 20-year-old female patient presented to an emergency outpatient department after sustaining a penetrating stab injury in the left upper chest area and was scheduled for an emergency thoracotomy to her left lung. The patient was intubated in the supine position using a left-sided double-lumen tube and turned to the left lateral position. The patient is 160cm tall.
What is the most likely depth of double-lumen tube insertion?
- A. 24 centimeters
 - B. 28 centimeters
 - C. 32 centimeters
 - D. 36 centimeters
- ANSWER: B
35. A 60-year-old male patient presented to an emergency department with the diagnosis of left upper lobe lung fibrosis and was scheduled for left lower lung lobectomy. Upon preoperative patient evaluation, the pulmonary function test revealed a forced expiratory volume in one second of 50 percent, oxygen consumption of 15 ml/kg/min, arterial blood pressure of oxygen of 60mm of mercury, and arterial blood pressure of carbon dioxide of 35mm of mercury. (The percentage of the right lower lung is 24 percent out of the total lung volume.).
What is the predicted postoperative pulmonary function in this case?
- A. 36 percent
 - B. 46 percent
 - C. 56 percent
 - D. 66 percent
- ANSWER: B
36. A 30-year-old known asthmatic female patient was brought to an emergency outpatient department with a complaint of an upper right chest penetrating bullet injury and was scheduled for emergency thoracotomy for right lung resection. During preoperative evaluation, the patient has been taking salbutamol for the last four years, and the chest was clear but has had a dry cough for the last three weeks. The patient underwent general anesthesia and was intubated using double-lumen tube. During intubation, the patient's oxygen saturation was 95 percent. Thirty minutes into the operation, the oxygen saturation became 80 percent, the patient developed wheezing, had an increased peak airway pressure, and the capnograph trace showed a slowly rising waveform, and she was diagnosed with intraoperative bronchospasm.
What is the most appropriate immediate management for this patient?
- A. Increase MAC of inhalational anesthetic to increase depth
 - B. Administration of aerosolized beta-2 agonists
 - C. Convert the DLT to a single tracheal tube
 - D. Turn off all anesthetic agents and ventilate the patient with 100 percent O₂
- ANSWER: B

37. A 37-year-old male patient is transferred to post-anesthesia care unit after emergency thoracotomy surgery under general anesthesia. Monitor reading shows SpO₂ of 88 percent, and paradoxical, inadequate chest wall movement with inspiratory stridor was noted. What is the most appropriate initial step in managing this patient?
- A. Re-intubation
 - B. Small dose of succinylcholine
 - C. Positive pressure ventilation
 - D. Sub-dose of intravenous propofol
- ANSWER: C
38. A 46-year-old male patient is admitted to a post-anesthesia care unit after emergency thoracotomy surgery under general anesthesia. On arrival, he is fully awake with vital signs within normal limits. A few minutes later, he started to complain of ipsilateral shoulder pain. What is the most appropriate management for this case?
- A. Gabapentin
 - B. Acetaminophen
 - C. Intercostal nerve block
 - D. Paravertebral block
- ANSWER: B
39. A 30-year-old male patient presents to a hospital with a history of anterior neck swelling. He is diagnosed to have controlled multinodular goiter and referred to an anesthetic clinic. On airway examination, only the soft palate and upper portion the uvula are visible. What is the most likely Mallampati class for this patient?
- A. Mallampati I
 - B. Mallampati II
 - C. Mallampati III
 - D. Mallampati IV
- ANSWER: B
40. A 40-year-old woman was scheduled for thyroidectomy. She has been on anti-thyroid drugs for the last six weeks. She has a history of heat intolerance, palpitation, and weight loss. Laboratory results show TSH of 0.2μIU/mL, T₃ of 180ng/dL, T₄ of 20μg/dL, serum ionized calcium of 3mEq/L, and serum phosphate of 1mmol/L. What is the most likely clinical interpretation for this case?
- A. Hypothyroidism
 - B. Hyperparathyroidism
 - C. Hyperthyroidism
 - D. Hypoparathyroidism
- ANSWER: C
41. During the surgical procedure of a 25-year-old woman, who is a known HIV/AIDS patient, the anesthesia team addresses the issue of a laryngoscope blade. There is a scarcity of this instrument to discard it. What is the most appropriate initial step of the laryngoscope blade sterilization strategy in this case?
- A. Blades may undergo high-level disinfection.
 - B. Placing the handle and blade on a clean surface separately.
 - C. The blade is rinsed and immersed in water with an enzymatic detergent.
 - D. Cross-infection arising from contaminated laryngoscopes is rare.
- ANSWER: C

42. A six-year-old male child with a history of obstructive sleep apnea (OSA) was brought to an operating theater to undergo an appendectomy. The anesthetist planned to provide opioid-free analgesia for this child.

What is the most likely reason to avoid opioid-based analgesia in this scenario?

- A. Increased sensitivity to respiratory depression
- B. Increased sensitivity to cardiovascular depression
- C. Aggravate postoperative urine retention
- D. Afraid of postoperative nausea and vomiting

ANSWER: A

43. A 40-year-old woman was scheduled for cholecystectomy. She has a history of malaria infection and has been on quinine and artesunate for the last two weeks. Physical examination reveals a blood pressure of 100/70mmHg, a heart rate of 95/min and an SPO₂ of 95 percent. The anesthetist on duty planned to maintain anesthesia with isoflurane and vecuronium.

What is the expected antimalarial drug-anesthesia drug interaction in this case?

- A. Quinine potentiates the action of NDMRs
- B. Quinine inhibits the action of NDMRs
- C. Artesunate potentiates the action of on NMDRs
- D. Quinine delays the onset of action of NDMRs

ANSWER: A

44. A 40-year-old woman was brought to an operating theater for left lung pleural decortication. The anesthetist on duty intubated her with double-lumen tube after induction with adequate muscle relaxation.

What is the most likely reason to use double-lumen tube in this scenario?

- A. To prevent contamination of healthy lung
- B. To control the distribution of ventilation to one lung
- C. To ventilate both lungs through the surgery
- D. To improve visualization of lung anatomy

ANSWER: A

45. A 45-year-old woman presented to a pre-anesthetic clinic after being scheduled for a total abdominal hysterectomy. She has a history of hyperparathyroidism secondary to parathyroid adenoma. On history, she reveals a history of anorexia, nausea, vomiting, weakness, and polyuria. Physical examination showed a blood pressure of 110/70mmHg, a heart rate of 95/min, and an SPO₂ of 95 percent. Laboratory findings show serum sodium of 140mEq/l, serum potassium of 3.8mEq/l, serum calcium of 10.5mg/dl, and serum chlorine ion of 102mEq/L. ECG revealed a shortened ST-segment and QT-interval.

What is the most likely electrolyte abnormality in this case?

- A. Hypokalemia
- B. Hyponatremia
- C. Hypercalcemia
- D. Hyperkalemia

ANSWER: C

46. A 35-year-old woman underwent a parathyroidectomy procedure for a parathyroid tumor. A day after the surgery, her blood pressure had dropped from 110/70mmHg to 70/50mmHg. She also became apneic. Postoperative serum electrolyte results revealed that she has hypocalcemia. What is the most appropriate initial management for this scenario?

- A. Infusion of 10 mL of 10 percent calcium gluconate
- B. Ordering of oral calcium supplements
- C. Supplementing with dihydrotachysterol
- D. Supplementing with calcium salts

ANSWER: C

47. During a surgical procedure of a 50-year-old woman, the anesthesia team wants to calculate the volume of oxygen in the E-cylinder for a 2-hour procedure. FIO₂ (inspired oxygen concentration) is set at 0.4 and the pressure in the cylinder is 1000psig. The flow rate is set at 2 liters/minute. What is the required volume of oxygen for the above-mentioned case?
- A. 240L
 - B. 340L
 - C. 330L
 - D. 165L
- ANSWER: A
48. A 30-year-old woman was scheduled for total thyroidectomy. She has been on propylthiouracil and propranolol for the last six weeks and she was confirmed to be clinically euthyroid. On preoperative evaluation, necessary laboratory parameters are ordered to ensure optimal anesthesia care. What action should be taken regarding this patient's anti-thyroid medication?
- A. It should be discontinued since the patient is clinically euthyroid
 - B. It should be continued until the morning of the surgery
 - C. Their usual dose should be doubled
 - D. Half of the usual dose should be taken
- ANSWER: B
49. A 45-year-old male patient was brought to an operating theater for emergency laparotomy due to perforated peptic ulcer disease. He has a history of gastroesophageal reflux disease, for which he experienced heartburn symptoms two times per week. What preoperative intervention is most likely advised for this patient?
- A. Planning to proceed with surgery with a supraglottic airway device
 - B. Premedication with non-particulate antacid is paramount
 - C. Premedication with sedative-hypnotics in the ward is recommended
 - D. Planning to proceed with a routine airway management strategy
- ANSWER: B
50. A 35-year-old woman presented to a pre-anesthetic clinic after being scheduled for nephrectomy due to renal cancer. She has no history of chronic medical illness. On physical examination, her blood pressure is 110/70mmHg, heart rate is 75/min, and SpO₂ is 95 percent. The anesthetist on duty ordered serum electrolytes before admission. What is the most likely reason for knowing the serum electrolyte of this patient?
- A. It is the component of routine laboratory parameters in this case
 - B. It is always in the normal range in this case
 - C. It is the only laboratory parameter to determine the overall patient health condition
 - D. Anesthetists should order it for every patient visit preoperative clinic
- ANSWER: A
51. A 70-year-old male patient came to a pre-anesthetic clinic after he was scheduled for right knee arthroplasty. He had a history of venous thromboembolism two months ago. Laboratory investigation shows an INR of 2.0, and prothrombin time is prolonged by 5 seconds. What is the most appropriate deep vein thrombosis prophylaxis plan in this case?
- A. Warfarin treatment should be continued until the night before surgery
 - B. LMWH should be continued until the morning of surgery if neuraxial block is planned
 - C. Ordering the patient to take bed rest for five days is warranted
 - D. Postponing the surgery in the first month is beneficial
- ANSWER: D

52. A 27-year-old male patient, who was diagnosed with small bowel obstruction, came to the operating theater for emergency laparotomy. The patient's blood pressure and heart rate were 80/60 mm Hg and 108/min respectively.

What is the best IV anesthetic agent for induction in this patient?

- A. Ketamine
- B. Propofol
- C. Thiopental
- D. Etomidate

ANSWER: A

53. A 48-year-old known asthmatic patient was scheduled for cholecystectomy. The patient's blood pressure is 100/65mmHg and heart rate is 108/min. During laboratory investigation, significant elevation of liver enzymes was observed.

What is the best choice of muscle relaxant to be used for maintenance in this case?

- A. Rocuronium
- B. Vecuronium
- C. Cisatracurium
- D. Pancuronium

ANSWER: C

54. A three-year-old male child is brought to an operating room for tonsillectomy. As the child was cooperative and vital signs were stable, the anesthetist decided to proceed with inhalational induction.

What is the best inhalational agent for this child?

- A. Isoflurane
- B. Halothane
- C. Sevoflurane
- D. Desflurane

ANSWER: C

55. A 70-kg relatively healthy woman, without any specific drug contraindications, came to an operating room for major orthopedic surgery. The senior anesthetist on duty formulated a multimodal analgesia plan.

What is the most appropriate pain management plan consideration in this case?

- A. Use of potent single analgesic medication targeted to one receptor
- B. Use of different medications that act on one receptor
- C. Use of a number of medication classes that act on different receptors
- D. Use of standard doses of multiple drugs that act on different receptors

ANSWER: C

56. An 85-kg woman, who had fasted for eight hours, came to an operating room for cholecystectomy. She has a preoperative hematocrit of 42 percent, and blood loss was 250ml.

What is the amount of maintenance fluid required for this woman?

- A. 125ml
- B. 175ml
- C. 225ml
- D. 255ml

ANSWER: A

57. A 39-year-old woman, a known asthmatic patient, came to an operating room for a total thyroidectomy. After she was intubated, she was struggling to reach the ETT with her hands, and she immediately developed wheezing. There was a prolonged expiratory phase on the capnograph and increased peak pressure on the ventilator.

What is the most appropriate initial management for this patient?

- A. Increase anesthesia depth
- B. Give salbutamol inhalation
- C. Give intravenous adrenaline
- D. Re-intubate the patient

ANSWER: A

58. A 57-year-old male patient who was diagnosed with large bowel obstruction was brought to an operating theater for emergency laparotomy. His blood pressure is 125/85mmHg and his heart rate is 96/min.

What is the best preventive measure for aspiration in this case?

- A. Insert nasogastric tube
- B. Administration of anti-acid
- C. Apply cricoid pressure
- D. Head up position

ANSWER: A

59. A 27-year-old male patient, who was diagnosed with an extensive third-degree burn of one week duration, came to an operating theater for debridement. His blood pressure is 125/85mmHg and his heart rate is 96/min.

What is the best muscle relaxant to be used during induction in this case?

- A. Succinylcholine
- B. Vecuronium
- C. Rocuronium
- D. Atracurium

ANSWER: C

60. A 26-year-old mother, who underwent cesarean section under general anesthesia, received a TAP block using a blind technique for postoperative analgesia.

What is the best indicator used to confirm appropriate nerve block in this case?

- A. Bounce of the needle by the internal oblique muscle
- B. Absence of blood during aspiration
- C. Loss of resistance during needle insertion
- D. Back flow of injected local anesthetics

ANSWER: D

61. A 35-year-old female patient, who underwent left-side abdominal wall hernia repair, received a Transversus Abdominis Plane Block (TAP) with a landmark technique as a component of multimodal postoperative analgesia.

What is the most appropriate anatomic location for local anesthetic administration in this case?

- A. Between internal oblique and external oblique
- B. Between transversus abdominis and internal oblique
- C. Between rectus abdominis and transvers
- D. Between latissimus dorsi and external oblique

ANSWER: B

62. A 55-year-old female patient, who was diagnosed with breast cancer, was brought to an operating room for a mastectomy. Initial intubation was attempted with both ETT and LMA but was not successful. Then, the anesthetist attempted bag mask ventilation, but encountered significant resistance, and the oxygen saturation dropped to 55 percent immediately.

What is the most appropriate next step in this patient's care?

- A. Another intubation attempt by senior
- B. Use of supraglottic airway
- C. Facemask ventilation with paralysis
- D. Emergency front of neck access

ANSWER: C

63. A 42-year-old male patient, who was diagnosed with primary hyperparathyroidism, came to an operating room for parathyroidectomy. On history taking, the patient was somnolent and had a severe muscle weakness. Laboratory results show a serum calcium concentration of 6.5mEq/L and an ionized calcium concentration of 3.5mEq/L. ECG reveals prolonged PR intervals and short QT intervals.

What is the most appropriate anesthesia technique for this patient?

- A. Increase the initial dose of depolarizing muscle relaxant
- B. Increase the initial dose of both depolarizing and non-depolarizing muscle relaxants
- C. Use the standard dose of both depolarizing and non-depolarizing muscle relaxant
- D. Decrease the initial dose of both depolarizing and non-depolarizing muscle relaxant

ANSWER: D

64. A 42-year-old male patient, who was diagnosed with obstructive jaundice, is scheduled for hepatobiliary surgery. On history taking, he has yellowish discoloration of the eyes. Then the anesthetist wanted to order a liver function test to confirm the diagnosis preoperatively.

What is the most appropriate investigation required for this patient?

- A. Serum Albumin
- B. Aspartate transaminase (AST)
- C. Alanine aminotransferase
- D. Alkaline phosphatase (ALP)

ANSWER: D

65. A 32-year-old morbidly obese woman was scheduled for adjustable gastric banding. During laryngoscopy, the patient moved her hands even though standard doses of drugs were used.

What is the most appropriate anesthetic management plan for this woman?

- A. Increase succinylcholine dose
- B. Increase propofol dose
- C. Increase halothane dose
- D. Increase morphine dose

ANSWER: A

66. A 64-year-old female patient, who was diagnosed with co-existing renal disease, came to an operating room for abdominal vascular surgery. The patient has blood pressure of 90/65mmHg, heart rate of 103/min and Sao2 of 99 percent at 1.5 L/min of gas flow.

What is the most likely inhalational agent to be avoided in this patient?

- A. Halothane
- B. Sevoflurane
- C. Desflurane
- D. Isoflurane

ANSWER: B

67. A 46-year-old male patient, who was diagnosed with complicated malaria, presented to an operating room for emergency laparotomy. He has a history of confusion and has started anti-malarial drug treatment.
What is the most appropriate anesthesia care plan for this patient?
- A. Avoid sedative drugs as premedication
 - B. Administer insulin to control hyperglycemia
 - C. Delay surgery until the patient responds to anti-malarial treatment
 - D. Administer ketamine and halothane during induction and maintenance
- ANSWER: A
68. A 29-year-old female HIV-seropositive patient came to an operating room for perineal abscess drainage under general anesthesia. The patient has a history of chronic cough of one month duration, which was not responsive to anti-TB medications.
What is the most appropriate strategy to prevent cross-transmission of infection from this patient?
- A. Wearing surgical gloves
 - B. Wearing an apron
 - C. Wearing an N-95 face mask
 - D. Wearing a surgical face mask
- ANSWER: C
69. A 52-year-old female HIV-seropositive patient with multiple neuropathies is brought to an operating room for perineal abscess drainage.
What is the most appropriate anesthesia technique for this patient?
- A. Epidural anesthesia
 - B. Spinal anesthesia
 - C. General anesthesia
 - D. Sciatic nerve block
- ANSWER: C
70. During the surgical procedure of a 45-year-old woman, who has a known history of controlled hypertension, the anesthesia team discussed her extubation plans at the end of the surgery. The anesthesia team confirmed that both subjective and objective criteria had been met by the patient.
What is the most appropriate extubation plan in this case?
- A. Awake extubation after confirming airway reflex intactness
 - B. Exchanging an ETT for an SGA while the patient is under deep anesthesia
 - C. Exchanging an ETT for an SGA while the patient is awake under anesthesia
 - D. Extubating during a light plane of anesthesia (stage II)
- ANSWER: B
71. A 60-year-old woman underwent a left lung decortication procedure. The anesthesia team confirmed the stability of all vital signs and the readiness of the patient for extubation. After extubation, the patient was transferred to a post-anesthesia care unit (PACU), and the anesthetist informed the PACU nurse about the patient.
What is the most likely vital information for the above-mentioned case?
- A. Particular attention should be given initiation of postoperative hydration
 - B. Particular attention should be given to monitoring oxygenation and ventilation
 - C. Particular attention should be given to monitoring neurologic function
 - D. Postoperative pain management is not a PACU nurse's concern and should be left to the anesthesia team
- ANSWER: B

72. A 35-year-old woman has undergone abdominal surgery for intestinal obstruction. After extubation, she became tachypneic and hypertensive. Postoperative chest radiography may demonstrate evidence of aspiration pneumonitis.

What is the most appropriate initial step to manage the above case?

- A. Delivery of supplemental oxygen and positive end-expiratory pressure
- B. Prophylactic antibiotics decrease the incidence of pulmonary infection
- C. Corticosteroid treatment of aspiration pneumonitis remains vital
- D. Measurement of tracheal aspirated pH and attempt to decrease it

ANSWER: A

73. A 35-year-old woman has undergone total abdominal hysterectomy. The surgical and anesthesia team took her to a post-anesthesia unit. Five minutes later, she complained of moderate pain. The anesthesia team discussed and initiated multi-modal analgesia regimens immediately.

What is the most likely reason that multimodal analgesia is preferred for acute postoperative pain in this case?

- A. To minimize opioid-induced adverse effects
- B. To increase opioid requirements along with opioid-sparing interventions
- C. To increase the chance of chronic post-surgical pain occurrence
- D. To decrease the requirements of local anesthesia infiltration

ANSWER: A

74. A 45-year-old woman underwent modified radical mastectomy. The surgical and anesthesia team took her to a recovery room. The patient's acute pain was managed by 2mg of morphine intravenously. The recovery nurse starts to monitor the patient's vital signs. Staying 10 minutes in the recovery room, the patient's vitals became stable.

What is the most appropriate time to discharge this patient to the ward?

- A. After 25 minutes
- B. After 50 minutes
- C. After 10 minutes
- D. After 15 minutes

ANSWER: D

75. A 65-year-old male patient is scheduled for a craniotomy. He has had stage-II hypertension for the last three years. He is on nifedipine and hydrochlorothiazide diuretics.

What is the most likely effect of the above scenario on cerebral physiology?

- A. Right shifting of auto-regulation curve
- B. Left shifting of auto-regulation curve
- C. Flow metabolism decoupling
- D. Flow metabolism coupling

ANSWER: A

76. A 28-year-old female patient is in an emergency department after sustaining traumatic brain injury. On neurologic assessment, she opens her eyes, shows abnormal flexion to pain and moans. Her vital signs are blood pressure of 160/95 mmHg, heart rate of 55/min, and SpO₂ of 90 percent. What is the most appropriate initial preoperative airway management for this case?

- A. Face mask ventilation
- B. Endotracheal intubation
- C. Laryngeal mask airway insertion
- D. Oro-pharyngeal airway insertion

ANSWER: B

77. A 30-year-old female patient is in an emergency department after sustaining traumatic brain injury. On neurologic assessment, she opens her eyes to pain stimuli and shows abnormal flexion to pain and makes incomprehensible sounds.
What is the most likely level of head injury for this case?
- A. Mild
 - B. Moderate
 - C. Severe
 - D. Very severe
- ANSWER: C
78. A 26-year-old male patient came to an operating theater for evacuation of subdural hematoma secondary to traumatic brain injury after a fall-down accident. He has a lower extremity injury that required bone stabilization. Related to the injury, he had lost 20 percent of his blood volume. He has two vomiting episodes preoperatively and complains of a severe headache. On physical examination, his GCS is 10, and his blood pressure is 80/55mmHg and pulse rate is 73/min.
What is the most preferred induction agent for this case?
- A. Propofol
 - B. Ketamine
 - C. Thiopental
 - D. Etomidate
- ANSWER: D
79. A 45-year-old male patient, who is diagnosed with a head injury, presented to the operating room for decompressive craniotomy. His Glasgow Coma Scale (GCS) score is 8 on admission. On arrival to the operation room, blood pressure of 180/105 mmHg, pulse rate of 55/min, and oxygen saturation of 85 percent under nasal oxygen were noted.
What is the most likely anesthesia management for this patient?
- A. Give 5 percent dextrose in normal saline
 - B. Tilt head down approximately 30 degrees
 - C. Administer mannitol before the dura is opened
 - D. Administer ketamine during induction of anesthesia
- ANSWER: C
80. A 55-year-old female patient with a history of hypertension is scheduled for elective neurosurgery. The anesthetist decided to apply induced controlled hypotension technique as a management plan during the intraoperative time.
What is the primary goal of the anesthetist in the above case?
- A. Speed up recovery after surgery
 - B. Prevent infection
 - C. Minimize bleeding
 - D. Improve relaxation
- ANSWER: C
81. A 50-year-old woman with a symptomatic 3-mm aneurysm of the bifurcation of the internal carotid artery presented for craniotomy and elective clipping of the aneurysm. Her blood pressure was 110/70mmHg, pulse rate was 70/min, and oxygen saturation was 99 percent. The surgical team needs the patient on deliberate hypotension.
What is the rationale for using induced hypotension in this case?
- A. Prevent rupture and bleeding of cerebral aneurysms
 - B. Treatment for cerebral vasospasm
 - C. It cannot cause acute renal failure
 - D. It has not been associated with visual disturbances
- ANSWER: A

82. A 36-year-old male patient, who sustained a head injury, came to an operating theater for an emergency craniotomy. A quick review of the chart showed that he has no documented medical illness. The vital signs were blood pressure of 90/50mmHg, pulse rate of 117/ min, oxygen saturation of 95 percent with a face mask under 100 percent oxygen. Laboratory findings reveal hemoglobin of 6.5g/dl, hematocrit of 20 percent, and platelet count of 180,000/ μ l.

What is the most appropriate course of action in this patient care?

- A. Administration of mannitol infusion
- B. Administration of noradrenaline infusion
- C. Administration of blood transfusion
- D. Administration of 5 percent DW

ANSWER: C

83. A 50-year-old, 75 kg female patient with a history of asthma is scheduled for posterior fossa surgery, and she has fasted for the last eight hours. Preoperative vital signs were blood pressure of 125/85mmHg, and pulse rate of 65/min.

What is the calculated fluid deficit of this patient?

- A. 880 milliliters
- B. 840 milliliters
- C. 720 milliliters
- D. 650 milliliters

ANSWER: B

84. A 45-year-old male patient with a history of frequent headache and neurologic deficit is scheduled for posterior fossa surgery in sitting position.

What is the most likely complication related to the patient's positioning?

- A. Intracranial hemorrhage
- B. Venous air embolism
- C. Cerebellar edema
- D. Arterial thrombosis

ANSWER: B

85. A 40-year-old man is brought to an emergency department after sustaining a severe closed head injury in a motor vehicle accident. His Glasgow Coma Scale (GCS) score is 7 on admission later intubated. There is a history of nausea, vomiting, and severe headache before deteriorating. The oxygen saturation is 90 percent to 92 percent on a face mask with 100 percent oxygen, blood pressure is 145/90mmHg, heart rate is 65/min, and respiratory rate is 24/min. The patient underwent depressed skull elevation and was transferred to the ICU. The anesthetist hands over the patient with strict postoperative management notes.

What is the acceptable target of postoperative ICP?

- A. < 15 mmHg
- B. 15-20 mmHg
- C. 20-30 mmHg
- D. 30-40 mmHg

ANSWER: A

86. A 28-year-old male patient was brought to a hospital with the diagnosis of primary traumatic brain injury after sustaining motorbike accident. Physical examination shows a GCS of 8. Vital signs were blood pressure of 90/60mmHg, heart rate of 95/min, and SpO₂ of 90 percent. Laboratory investigation shows a hematocrit level of 39 percent. ABG (PCO₂ 35mmHg and PO₂ 80mmHg). Decompressive craniectomy under general anesthesia with ETT was performed, and on emergence from anesthesia, vital signs became blood pressure of 80/55mmHg, heart rate of 75/min, SpO₂ of 98 percent on oxygen, and temperature of 35 degrees Celsius.

What is the most appropriate postoperative airway management plan for this patient?

- A. Extubate the patient and ensure rewarming in the operation theatre
- B. Blunt hypertensive response to emergence and extubation by IV lidocaine
- C. Avoidance of coughing before extubation and control of systemic BP
- D. Mechanically ventilate postoperatively and extubate after careful rewarming in the ICU

ANSWER: D

87. A 24-year-old male patient came to a hospital for preoperative anesthesia evaluation for a tonsillectomy procedure. He reported that he snores loudly at nighttime and he feels tired and fatigued during the daytime. He did not take any medications, and his vital signs were in the normal range.

What is the most likely risk of this patient for obstructive sleep apnea?

- A. Low Risk
- B. Mild Risk
- C. Moderate Risk
- D. High Risk

ANSWER: D

88. A three-year-old male child presented to an operation room due to difficulty of breathing. He is infected with Hemophilus influenzae. He has a sore throat, dysphagia, and a muffled voice. To ease the breathing, he prefers the “tripod” position.

What is the most appropriate airway management option for this case?

- A. LMA insertion
- B. Bag-Mask Ventilation
- C. Oral endotracheal intubation
- D. Awake Nasal Endotracheal Intubation

ANSWER: C

89. A 56-year-old male patient is scheduled for vitrectomy with general anesthesia and endotracheal intubation. He has no pertinent history of past medical illness. Preoperatively, his blood pressure and heart rate have increased by 20 percent from the baseline, and he is restless and sweating.

What is the most likely directly affected physiologic factor to increase intraocular pressure in this case?

- A. Scleral Volumes
- B. Aqueous Volumes
- C. Vitreous Volumes
- D. Choroidal Volumes

ANSWER: D

90. A 59-year-old woman presented to an operating theater for elective cholecystectomy on a day-case basis. She is a nonsmoker and has obstructive sleep apnea and well-controlled hypertension. Her vital signs are within the normal range and body mass index is 35kg/m².

What is the most beneficial premedication for this case?

- A. Diazepam
- B. Morphine
- C. Furosemide
- D. Dexamethasone

ANSWER: D

91. A two-year-old male child is under general anesthesia with LMA for strabismus repair. The child has no other pertinent systemic illnesses and vital signs are within the normal range. Induction was uneventful. Twenty minutes after the initiation of the surgery, his blood pressure became 70/30 mmHg, his heart rate became 42/min, and SpO₂ became 89 percent. What is the most appropriate initial course of action for this case?
- A. To administer Atropine
 - B. To administer Epinephrine
 - C. To massage the carotid artery
 - D. To stop surgical procedures
- ANSWER: D
92. A two-year-old male child presented to an operation room with foreign body aspiration located in the laryngeal area. He is suffering from difficulty of breathing. It is planned to induce the child with an inhalational agent. What is the most preferable induction position for this child?
- A. Sitting
 - B. Supine
 - C. Lateral
 - D. Trendelenburg
- ANSWER: A
93. A 14-year-old female patient presented to an operation theater for abscess drainage from the floor of the mouth. The abscess has involved the submandibular, sublingual, submental, and buccal spaces. The infection has resulted in trismus. Due to breathing difficulties, she avoids the supine position. What is the most appropriate initial airway management option for this case?
- A. Nasal cannula
 - B. Laryngeal Mask Airway
 - C. Needle cricothyrotomy
 - D. Awake fiber optic intubation
- ANSWER: D
94. A four-year-old female child is under general anesthesia with endotracheal intubation for adenotonsillectomy. The induction was uneventful, and she is breathing spontaneously. During the surgery, the anesthesia machine starts to alarm for a PIP of 35cmH₂O. When the chest is auscultated, there is decreased air entry but no added sounds. There is a progressive decrease in the capnograph trace. What is the most likely cause of the alarm in this case?
- A. Kinked endotracheal tube
 - B. Dislodged endotracheal tube
 - C. Migrated endotracheal tube to main bronchus
 - D. Disconnection between circuit and endotracheal tube
- ANSWER: A
95. A 54-year-old male patient presented to an operation theater for resection of naso-pharyngeal angiofibroma. He has hypertension, and his ECG findings show an elevated ST segment on the V5 and V6 leads. He is induced and intubated with an endotracheal tube. During surgery, there is massive bleeding from the surgical site. What course of action most likely would most likely have prevented the complication?
- A. Head-up positioning
 - B. Controlled hypotension
 - C. Topical application of cocaine
 - D. Topical application of lidocaine with epinephrine
- ANSWER: A

96. A seven-year-old child presented to an operating room with an open eye injury. His mother reported that the injury took place two hours ago and he had eaten solid meal four hours ago. What is the most preferred anesthesia management for this case?

- A. Regional anesthesia
- B. Monitored anesthesia care
- C. General anesthesia with LMA
- D. General anesthesia with ETT

ANSWER: D

97. A three-year-old male child is in the PACU after a tonsillectomy procedure. The surgery and anesthesia management were uneventful. Four hours after surgery, the child becomes restless and tries to cry but loses strength. On physical examination, he is pale with some blood spots observed around the mouth. His vital signs are blood pressure of 70/40mmHg, heart rate of 120/min, and SPO₂ of 84 percent. The surgeon decided to re-operate.

What is the most appropriate initial course of action in managing this case?

- A. Fluid resuscitation
- B. Oro-pharyngeal suctioning
- C. Rapid sequence induction
- D. Naso-pharyngeal airway insertion

ANSWER: A

98. A six-year-old male child developed inspiratory stridor immediately after awake endotracheal extubation. The performed procedure was total tonsillectomy. His heart rate is 93/min and SPO₂ is 97 percent. As soon as the anesthetist noticed this, continuous positive airway ventilation (CPAP) with 20cmH₂O is initiated. After one minute of CPAP, the intensity and volume of the stridor increased, and auscultation confirmed no air entry. His vital signs became a heart rate of 73/min and a SpO₂ of 86 percent.

What is the most appropriate medication to manage this case's condition?

- A. Atropine
- B. Ketamine
- C. Lidocaine
- D. Succinylcholine

ANSWER: D

99. A 77-year-old male known type-II diabetic patient for the last 10 years came to a preoperative clinic after he was scheduled for cholecystectomy. Preoperative vital signs show blood pressure of 125/80mmHg while sitting and 90/60mmHg while standing; a heart rate of 143/min and oxygen saturation of 85 percent under room air.

What is the most important investigation to be examined for this scenario?

- A. Liver function test
- B. Renal function test
- C. Chest x-ray
- D. ECG

ANSWER: D

100. A 65-year-old male known type-II diabetic patient for the last five years came to a preoperative clinic after he was scheduled for nephrectomy. He was on 500mg of metformin twice daily for the last five years. Preoperative laboratory findings reveal random blood sugar of 200mg/dl, hemoglobin level of 12g/dl, hematocrit of 36 percent, platelet count of 250,000/ μ l, and creatinine of 1.1mg/dl.

What is the most appropriate management plan for this patient care?

- A. Regular exercise for diabetic optimization
- B. Start 15 units of regular insulin infusion with normal saline
- C. Maintain intraoperative serum blood sugar below 180 mg/dl
- D. Put him on a sliding scale for the next one week before proceeding to surgery

ANSWER: C

101. A 45-year-old female patient with a history of suicide attempt two months ago came to a preoperative clinic for cholecystectomy. On history, it was known that she had been on tricyclic antidepressants for the last year.

What is the most likely relevant physical examination for the case scenario presented above?

- A. Identification of arrhythmia on palpation
- B. Identification of mass on chest percussion
- C. Identification of central cyanosis on inspection
- D. Identification of yellowish discoloration on inspection

ANSWER: A

102. A 46-year-old female patient with a history of bipolar disorder came to a preoperative clinic for elective nephrectomy. She has been on lithium for the last five years.

What is the most likely drug interaction in this patient?

- A. Lithium can increase anesthetic requirements
- B. Lithium induces sensitivity to non-depolarizing relaxants
- C. Lithium blood levels decrease with thiazide
- D. Lithium should be stopped before the morning of surgery

ANSWER: B

103. A 75-year-old male patient with a history of Parkinson's disease came to a preoperative clinic for a laparotomy. He has no history of other comorbidities and has been on levodopa for the last five years.

What is the most appropriate anesthesia management for this patient?

- A. Continue levodopa to the morning of surgery
- B. Discontinue levodopa to the morning of surgery
- C. Administer 0.6 mg of atropine before induction
- D. Administer 20 mg of metoclopramide before induction

ANSWER: A

104. A 40-year-old female patient with a history of hypertension came to an operating theater for nephrectomy. During induction, standard doses of IV lidocaine, fentanyl, propofol, and suxamethonium were used to intubate the patient. However, the patient's blood pressure and heart rate were rising significantly in the middle of the surgery, and 2% halothane was used to alleviate the signs.

What is the most appropriate extubation plan for this patient?

- A. Deep plane of anesthesia before extubation
- B. Light plane of anesthesia before extubation
- C. Administration of beta-blockers after extubation
- D. Suctioning of the patient after extubation

ANSWER: A

105. A 60-year-old female patient with a history of long-standing diabetes is scheduled for transurethral resection of the prostate under spinal anesthesia. During the immediate preoperative period, her vital signs are stable and urine output is adequate. Laboratory investigations show RBS of 350 mg/dl, potassium of 2.9 mEq/L, and pH of 7.2. Insulin infusion and isotonic crystalloid fluid were started after the serum sugar was confirmed.

What is the most appropriate next step during resuscitation of this patient?

- A. Administration of sodium bicarbonate
- B. Administration of insulin bolus
- C. Administration of potassium
- D. Administration of glucose

ANSWER: C

106. A 46-year-old male patient with a history of depression, type-II diabetes, and hypertension came to a preoperative clinic for elective craniotomy. He is on monoamine oxidase inhibitors, metformin, and nifedipine.

What is the most appropriate anesthesia management for this patient?

- A. Decrease the anesthetic dose
- B. Increase the anesthetic dose
- C. Use ephedrine to treat hypotension
- D. Stop MAOIs before elective surgery

ANSWER: B

107. An 87-year-old female patient with a history of anxiety underwent emergent exploratory laparotomy. Her vital signs show a blood pressure of 130/ 70 mmHg, a heart rate of 80/min, and an oxygen saturation of 87 percent under room air. Upon successful accomplishment of surgery, the patient is transferred to an intensive care unit, and she attempted to remove the endotracheal tube and urinary catheter herself.

What is the most likely level of postoperative delirium for this patient?

- A. Combative
- B. Agitated
- C. Very agitated
- D. Restless

ANSWER: C

108. An 86-year-old female patient with a history of dementia underwent emergency laparotomy. Vital signs show a blood pressure of 130/ 70 mmHg, a heart rate of 80/min, and an oxygen saturation of 87 percent under room air. The surgical team finished successfully, but in the recovery room the patient experienced difficulty of focusing attention and incoherent thinking.

What is the most appropriate postoperative plan for this patient?

- A. Administer a high dose pethidine for postoperative pain
- B. Administer a low dose ketamine IV infusion for sedation
- C. Administer antipsychotic medications at the lowest effective dose
- D. Administer anticholinergic medication at the lowest effective dose

ANSWER: C

109. A one-year-old male child is brought to an operating room for a colostomy. Upon preoperative assessment, both lungs are clear, and there is no history of cough, but the patient is restless. His vital signs show a blood pressure of 95/60 mmHg, a heart rate of 125/min, and a respiratory rate of 28/min. The anesthetist premedicates the patient with oral midazolam in combination with ketamine.

What is the most likely rationale for the premedication in the above scenario?

- A. To decrease pain
- B. To decrease anxiety
- C. To decrease secretion
- D. To decrease induction time

ANSWER: B

110. A two-year-old child with obstructive sleep apnea presented for tonsillectomy and adenoidectomy. The scheduled anesthetist does a physical examination and plans to intubate with a cuffed endotracheal tube.

What is the most appropriate internal diameter of the tube for this patient?

- A. 3 mm
- B. 4 mm
- C. 5 mm
- D. 6 mm

ANSWER: C

111. A five-year-old child is brought to an operation room after being diagnosed with intussusception. Resection and anastomosis were planned under general anesthesia with endotracheal intubation and later induced with intravenous ketamine and suxamethonium to facilitate endotracheal intubation. Vecuronium with sevoflurane is planned for maintenance.

What is the most efficient circuit for the above case?

- A. Mapleson A
- B. Mapleson B
- C. Mapleson C
- D. Mapleson D

ANSWER: D

112. A four-month-old male child is brought to a preoperative anesthesia clinic for evaluation before an elective surgery. Clinically, he has a severe cough with a runny nose and sore throat of two weeks duration. An axillary thermometer reveals fever of 39 degrees Celsius.

What is the most appropriate clinical decision for this patient?

- A. Proceed with the surgery if the chest is clear
- B. Reschedule the patient after four weeks
- C. Proceed with the surgery if there are no rhonchi
- D. Reschedule the patient after two days

ANSWER: B

113. An eight-week-old child presented to an emergency department with a history of intolerance to feedings. The patient has had nonbilious projectile vomiting immediately after meals for the past 48 hours. Laboratory investigations reveal potassium of 2.8 mEq/L, sodium of 120 mEq/L, chloride of 90 mEq/L, and bicarbonate of 35 mEq/L. Ultrasound result shows an olive-shaped mass in the right upper quadrant.

What is the most appropriate immediate management for this patient?

- A. Adequate resuscitation
- B. Order further investigations
- C. Aspirate with orogastric tube
- D. Surgical removal of the mass

ANSWER: A

114. A three-year-old male child is brought to an operation theater for an emergency orthopedic procedure of the upper extremity. Clinically, he has a runny nose with a severe cough and sore throat of three weeks duration. Axillary temperature is 38.5 degrees Celsius. The anesthetist plans to proceed with general anesthesia with LMA insertion.

What is the most appropriate inhalational drug for this patient?

- A. Thiopental with fentanyl
- B. Ketamine with fentanyl
- C. Halothane with fentanyl
- D. Sevoflurane with fentanyl

ANSWER: D

115. A 10-kg female child with an acute abdomen is brought to an operation room for an elective laparotomy. She has not received oral or intravenous intakes for four hours prior to the operation. What is the most likely fluid deficit in the first hour of surgery?

- A. 80 ml
- B. 90 ml
- C. 100 ml
- D. 120 ml

ANSWER: D

116. A four-year-old male child presented to a hospital after being diagnosed with adenoid hypertrophy and scheduled for tonsillectomy. After standard induction, the child was intubated with 5.5 mm ETT. Immediately after extubation, the child is straining, and there is audible inspiratory stridor, and SpO₂ is 88 percent. Later, the child is diagnosed with post-extubation croup.

What is the most effective management for this patient?

- A. Helix administration
- B. Intravenous dexamethasone
- C. Administration of intramuscular suxamethonium
- D. Racemic epinephrine administration

ANSWER: D

117. A two-year-old male child is brought to an operation room for an emergency laparotomy. During the surgical procedure, there is prolonged wound exposure despite being in an inadequately warmed room. In addition, he took intravenous fluid and dry anesthetic gases.

What is the most likely impact of the above-mentioned finding?

- A. Fast recovery with stable respiratory function
- B. Cardiac arrhythmia
- C. Pulmonary arterial hypotension
- D. Decreased pulmonary vascular resistance

ANSWER: B

118. A four-week-old child presented with projectile vomiting after feedings, dehydration, and failure to thrive. The vomiting is nonbilious and persistent for the past 48 hours. Laboratory findings show potassium of 2.5 mEq/L, bicarbonate of 35 mEq/L, and chloride of 76 mEq/L. Blood pressure of 90/50 mmHg, a heart rate of 130/min, and SpO₂ of 94 percent at room temperature were noted.

What is the most likely anesthetic management plan for this child?

- A. Delay surgery and resuscitate
- B. Proceed with surgery since it is an emergency
- C. Avoid insertion of NG tube to facilitate enhanced recovery
- D. Does not need muscle relaxation, hence it being a short procedure

ANSWER: A

119. A five-day-old male newborn baby with lower gastrointestinal tract obstruction is brought to an operation theater. Recent laboratory results revealed deterioration of electrolytes within a short period of time. The surgical team describes that it may be due to sequestration within the intestinal tract. The patient had been NPO for the last eight hours. The newborn's blood pressure is 90/50 mmHg, heart rate is 130/min, SpO₂ is 94 percent at room temperature, and temperature is 38.5 degrees Celsius.

What is the most appropriate anesthesia management plan?

- A. Rapid sequence induction is barely needed as the patient is fasted
- B. Cooling with inhalational anesthetics and muscle relaxants has no benefit
- C. Ensuring that fluid replacement is sufficient
- D. Extubate the trachea with the child in semi-supine and a deep level of anesthesia

ANSWER: C

120. A nine-month-old infant with hypernasal speech and nasal turbulence is scheduled for repair of a cleft palate. The infant was born full term and has no other medical problems. During preoperative assessment, there are other craniofacial anomalies or retrognathia. The patient was later diagnosed with cleft lip and palate.

What is the most appropriate technique to secure the airway of this patient?

- A. Endotracheal intubation with controlled ventilation
- B. Endotracheal intubation with spontaneous ventilation
- C. Laryngeal mask insertion with controlled ventilation
- D. Laryngeal mask insertion with spontaneous ventilation

ANSWER: B

121. An eight-year-old male child came to an operation theater for wound care and burn dressing. His last procedure was done under procedural sedation without complication, and he currently shows no signs of respiratory tract infection. The anesthetist plans to sedate the patient and begin the procedure promptly.

What is the most appropriate drug for this patient?

- A. Ketamine
- B. Diazepam
- C. Morphine
- D. Halothane

ANSWER: A

122. A two-year-old male child with intestinal obstruction presented to an operation room for an emergency procedure. After intubation, he developed a rapid pulse, and the cardiac monitor showed supraventricular tachycardia (SVT).

What is the most appropriate drug for this patient?

- A. Lidocaine
- B. Adenosine
- C. Amiodarone
- D. Procainamide

ANSWER: B

123. A four-year-old male child with an acute abdomen underwent an appendectomy under general anesthesia. He was induced with propofol and maintained with sevoflurane and mivacurium. After the completion of the procedure, the child has not regained consciousness. Currently, he has a blood pressure of 95/65 mmHg, a heart rate of 120/min, a respiratory rate of 26/min, an oxygen saturation of 97 percent, and a temperature of 32.6 degrees Celsius.

What is the most likely cause for the delayed emergence of this patient?

- A. Hypoxia
- B. Hypotension
- C. Hypothermia
- D. Drug toxicity

ANSWER: C

124. A five-year-old male child, who underwent upper abdominal surgery, has been in the post-anesthesia care unit for four hours. Initially, he had restlessness and tachycardia, but he is now obtunded, with bradycardia, and hypotension. The anesthetist examines the patient for further diagnosis and management. He has a heart rate of 75/min, and a SpO₂ of 85 percent.

What is the most appropriate management for this patient?

- A. Routine administration of oxygen through ETT
- B. Administration of oxygen through a mask
- C. Immediate administration of atropine and frequent observation
- D. Administration of aerosolized bronchodilator

ANSWER: B

125. A 32-year-old male patient presented to an emergency outpatient department two hours after a mass casualty incident in a road traffic accident. Upon clinical evaluation, he has right leg pain and paradoxical chest movement. He is able to open his eyes to speech, responds appropriately, and has abnormal extension of extremities.

What is the most likely correct Glasgow Coma Score of this patient?

- A. 7
- B. 8
- C. 9
- D. 10

ANSWER: B

126. A 30-year-old female patient was admitted to an ICU with a diagnosis of acute respiratory failure and sudden loss of consciousness three days ago. The patient developed hypotension, which is unresponsive despite adequate fluid resuscitation. Examination findings show a blood pressure of 80/45 mmHg, a heart rate of 130/min, a respiratory rate of 25/min, and a temperature of 38 degrees Celsius. She also has the clinical manifestation of lactic acidosis, oliguria, and alteration in mental status along with a Glasgow Coma Scale of 8/15.

What is the most likely diagnosis of this patient?

- A. Septic shock
- B. Cardiogenic shock
- C. Obstructive shock
- D. Spinal shock

ANSWER: A

127. A 25-year-old female patient presented to an emergency department with complaints of headache, nausea, and vomiting. During assessment, the patient is restless and confused. She is in distress and has difficulty of breathing; the family reported that she is at home with her family and has had no recent ingestion of any beverage or food. The patient has a blood pressure of 100/60 mmHg, a heart rate of 125/min, a respiratory rate of 22/min, an oxygen saturation of 85 percent at room air, and a temperature of 36 degrees Celsius.

What is the most likely diagnosis of this patient?

- A. Alcohol intoxication
- B. Hypoglycemic shock
- C. Organophosphate poisoning
- D. Carbon monoxide poisoning

ANSWER: D

128. A 32-year-old male patient presented to an emergency outpatient department after a mass casualty secondary to a road traffic accident. Upon clinical presentation, he has right leg pain and paradoxical chest movement. He is able to open his eyes to speech, is disoriented to TPP, and has abnormal extension of extremities. On examination, the patient has a blood pressure of 100/60 mmHg, a heart rate of 90/min, and a respiratory rate of 32/min. ABG result shows PaO₂ of 45 mmHG and PaCO₂ of 65 mmHG. The anesthesia team planned to admit the patient to the ICU and decided to put in a mechanical ventilator.

What is the most appropriate mode of mechanical ventilator to be used in this case?

- A. Pressure Support Ventilation
- B. Pressure-controlled Ventilation
- C. Intermittent Mandatory Ventilation
- D. Continuous Mandatory Ventilation

ANSWER: A

129. A 60-year-old known cardiac male patient is brought to an outpatient department after a fall-down accident and scheduled for internal fixation with a diagnosis of left arm fracture. Vital signs show a blood pressure of 130/80 mmHg, a heart rate of 85/min, and a SpO₂ of 90 percent on room air. The anesthesia team planned to proceed with general anesthesia. Intraoperatively, the patient took 2000 ml of maintenance fluid. Surgery is completed, and the patient is transferred to PACU. The patient developed frothy secretions and difficulty of breathing, and became restless 30 minutes postoperatively. On auscultation, there is crackle all over the chest. The oxygen saturation of the patient became 85 percent.

What is the most likely diagnosis of this patient?

- A. Laryngospasm
- B. Bronchospasm
- C. Pulmonary edema
- D. Pneumonia

ANSWER: C

130. A 74-year-old female patient with a hepatic head tumor had undergone hepatobiliary surgery. Intraoperatively, she had significant hemodynamic instability with hepato-renal failure. She needs respiratory, cardiovascular, renal, and hepatic organ support in the perioperative period.

Where is the most appropriate area to manage this patient during the postoperative period?

- A. Post-anesthesia care unit
- B. Intensive care unit
- C. Surgical ward
- D. Medical ward

ANSWER: B

131. A 48-year-old male patient came to a hospital with a complaint of pain in the perineum. He has nausea and vomiting but no trauma history or any drug abuse. On assessment, he is in shock with an increased pulse rate and pulse pressure; he has warm extremities with rapid capillary refill and a temperature of 38.7 degrees Celsius. He has a WBC count of 3500/mm³.

What is the most likely diagnosis of this patient?

- A. Cardiogenic shock
- B. Hypovolemic shock
- C. Septic shock
- D. Obstructive shock

ANSWER: C

132. A 37-year-old male patient is admitted to an ICU due to respiratory failure secondary to a severe asthmatic attack. He is on a mechanical ventilator with a pressure-controlled ventilatory mode. Ventilator parameters include a peak inspiratory pressure of 28cm H₂O and a respiratory rate of 14/min. An arterial blood gas analysis was performed to establish ventilator strategies, and the result shows PaCO₂ of 55 mmHg, PaO₂ of 78 mmHg, pH of 7.32, and HCO₃⁻ of 28mEq/L (Reference: PaCO₂ is 35-45 mmHg, PaO₂ is 70-100 mmHg, pH is 7.35-7.45, HCO₃⁻ is 24 mEq/L).

What is the most likely primary acid-base disturbance for this case?

- A. Metabolic Acidosis
- B. Metabolic Alkalosis
- C. Respiratory Acidosis
- D. Respiratory Alkalosis

ANSWER: C

133. A 44-year-old patient is admitted to an ICU with a chief complaint of difficulty of breathing. Upon investigation, PaO₂: FiO₂ is less than 200 with diffuse infiltrates on chest radiograph. The patient has a blood pressure of 90/70 mmHg, a heart rate of 120/min, a respiratory rate of 28/min, and an SpO₂ of 90 percent. The anesthetist in charge of the ICU decided to intubate and ventilate the patient.

What is the most likely initial mode of ventilation?

- A. Volume control ventilation (VC)
- B. Pressure control ventilation (PC)
- C. Intermittent mandatory ventilation
- D. High-frequency positive pressure ventilation

ANSWER: A

134. A 46-year-old man is rushed to an operating room for emergency laparotomy and thoracotomy after he sustained multiple abdominal and thoracic gunshot wounds. In the field, paramedics intubated the patient, began fluid resuscitation with two large-bore intravenous lines, and inflated a pneumatic anti-shock garment. Upon arrival in the operating room, the patient's pulse and blood pressure are unrecordable.

What is the immediate course of action for this patient?

- A. Blood transfusion
- B. Chest tube insertion
- C. Cardiopulmonary resuscitation
- D. Abdominal surgery

ANSWER: C

135. A 36-year-old female patient is in the emergency department after sustaining a traumatic brain injury. She is being evaluated for potential surgical intervention. On initial assessment, her Glasgow Coma Scale (GCS) score is 8, and she is intubated. Her vital signs include a blood pressure of 90/50 mmHg, a heart rate of 130/min, a respiratory rate of 20/min, and SpO₂ of 90 percent.

What is the most preferable initial fluid choice for optimizing this case?

- A. Mannitol
- B. 0.9 percent normal saline
- C. Albumin
- D. 5 percent dextrose

ANSWER: B

136. A 26-year-old G-II, P-I woman, at the 36th week of gestation, suddenly collapses and is unresponsive when reaching an emergency department. Her blood pressure is not measurable, and her pulse is not palpable.

What is the most appropriate management for this patient?

- A. Start chest compressions
- B. Administer adrenaline
- C. Start defibrillation
- D. Administer atropine

ANSWER: A

137. A 24-year-old male patient was brought to an emergency outpatient department one hour after sustaining a car accident. During patient assessment, the patient has chest pain, difficulty of breathing and no history of loss of consciousness. On physical examination, there is an open wound over the right side of the chest, distended neck veins, and tracheal deviation to the left. On auscultation, there is hyper-resonance and a decreased breath sound over the right side of the chest. His vital signs are blood pressure of 80/55 mmHg, heart rate of 120/min, respiratory rate of 30/min, and temperature of 36 degrees Celsius.

What is the most likely diagnosis of this patient?

- A. Tension pneumothorax
- B. Open Pneumothorax
- C. Massive hemothorax
- D. Cardiac Tamponade

ANSWER: A

138. A 30-year-old man is brought to an emergency outpatient department two hours after sustaining a motor vehicle accident. On initial assessment, he is responding to commands, and his respiration is normal with good air entry bilaterally. However, there is visible bleeding from lower extremities de-gloved soft tissue injury; there is a fracture of bone in the lower leg with visible deformity and swelling. The patient is irritable and in distress.

What is the most appropriate immediate intervention for this patient?

- A. Oxygen supplementation
- B. Apply a tourniquet to halt bleeding
- C. Application of Plaster Splinting
- D. Administer intravenous fluids with a large-bore cannula

ANSWER: C

139. A 34-year-old female patient is brought to an emergency outpatient department with a complaint of shortness of breath two hours after sustaining a penetrating injury to the left side of the chest. On arrival, the vital signs were blood pressure of 85/55 mmHg, heart rate of 125/min, and respiratory rate of 20/min. On auscultation, there is decreased breath sound on the left side of the chest. On inspection, there is an open wound over the right side of his chest, and auscultation revealed hyper-resonance over the left side of the chest. Needle decompression has been immediately done by the emergency resident.

What is the most appropriate immediate next-step intervention for this patient?

- A. Requesting ultrasound
- B. Performing emergency thoracotomy
- C. Requesting control Chest x-ray
- D. Inserting chest tube

ANSWER: D

140. An 18-year-old patient is brought to an emergency outpatient department by a traffic police officer three hours after suffering multiple thoraco-abdominal injuries secondary to a motor vehicle accident. On arrival, the patient is not responsive, his blood pressure is unrecordable, and no peripheral pulse is detected.

What is the most appropriate initial management for this patient?

- A. Immediate intubation and ventilation
- B. Immediate cardiopulmonary resuscitation
- C. Needle decompression and chest tube insertion
- D. IV line opening and crystalloid administration

ANSWER: B

141. An 85-kg woman with a preoperative hematocrit of 35 percent has been scheduled to undergo open reduction and internal fixation of a radio-ulnar fracture. The anesthetist plans to estimate the allowable blood loss to guide intraoperative transfusion decisions. (Use Estimated Blood Volume (EBV) is 65ml/kg).
What is the allowable blood loss amount to maintain her hematocrit at 30 percent?
A. 800 ml
B. 700 ml
C. 600 ml
D. 500 ml
ANSWER: A
142. A 35-year-old male patient is scheduled to operating room with a diagnosis of femoral shaft fracture secondary to a motor vehicle accident and is undergoing fixation of the femoral shaft fracture under general anesthesia. Following the administration of half a unit of packed red blood cells, the patient develops tachycardia, hypotension, and dyspnea. The surgical team immediately stops the transfusion and plans an alternative management plan.
What is the most appropriate next management for this patient?
A. Diphenhydramine
B. Epinephrine
C. Cimetidine
D. Aminophylline
ANSWER: B
143. A 50kg, 18-year-old female patient presented to an emergency outpatient department with burn injuries. During evaluation, she has thermal burn injuries to her lower legs and back. The total calculated total body surface area is 36 percent.
According to the Parkland formula, what is the correct amount of fluid required in the first 24 hours?
A. 4200 mL
B. 5200 mL
C. 6200 mL
D. 7200 mL
ANSWER: D
144. A 50-kg, 30-year-old female patient is brought to an emergency outpatient department with a complaint of thermal burn injury on her left leg, face, and anterior chest. On assessment, the burn involves the dermis, lymphatic channels, nerves, and blood vessels.
What is the most appropriate classification of this burn?
A. First-degree burn
B. Second-degree burn
C. Third-degree burn
D. Fourth-degree burn
ANSWER: C
145. A 55-kg, 40-year-old female patient is admitted to a surgical ward with a diagnosis of second-degree thermal burn injury involving her gluteal area and upper and lower extremities. The emergency physician transferred the patient to an operating room for debridement and skin graft on the 8th day after the burn. The anesthetist plans to proceed with general anestheisa under controlled ventilation.
What is the most appropriate anesthesia drug for this patient?
A. Diazepam
B. Ketamine
C. Morphine
D. Suxamethonium
ANSWER: B

146. A 65-year-old male patient, who was admitted to a surgical ward with a diagnosis of second-degree burn at the anterior chest and lower extremity area, is taken to an operating room for skin grafting under general anesthesia. The senior anesthetist advised the use of a non-depolarizing muscle relaxant (vecuronium) instead of a depolarizing muscle relaxant (Suxamethonium) along with 100 percent oxygen and positive pressure ventilation. What is the most likely explanation for this professional advice?
- A. Proliferation of extrajunctional neuromuscular receptors
 - B. Desensitization of post-junction nicotinic receptors
 - C. Increased reuptake of acetylcholine at synaptic cleft
 - D. Increased calcium channel depolarization
- ANSWER: A
147. A 20-year-old male patient is undergoing surgical intervention for compression of the spinal cord secondary to a tumor above T6 under general anesthesia. At the conclusion of surgery, while the patient is transferred to PACU, the blood pressure is constantly increased while the heart rate is decreased. The patient complains of a severe headache and blurring of vision. What is the most appropriate management of this patient?
- A. Ephedrine
 - B. Sodium nitroprusside
 - C. Morphine
 - D. Dobutamine
- ANSWER: B
148. A 24-year-old male patient with a spinal cord injury at T6 is brought to an operation room four days after a car accident. Emergency surgical management is undertaken to reverse compression of the spinal cord due to dislocation of a vertebral body. The anesthetist found that the patient has developed hypotension and severe bradycardia prior to any anesthetic administration and intubation. What is the most appropriate initial anesthesia management for this patient?
- A. Fluid resuscitation followed by ketamine anesthesia
 - B. Antibiotic support along with fluid resuscitation
 - C. Fluid support followed by propofol anesthesia
 - D. Corticosteroid support followed opioid-based anesthesia
- ANSWER: A
149. A 28-year-old male patient is undergoing a total knee arthroplasty procedure. A thigh tourniquet has been applied to minimize blood loss during the procedure. The surgical procedure has been uneventfully concluded, and the tourniquet is deflated after 120 minutes of use. What is the most likely clinical finding observed immediately after tourniquet release?
- A. Hypothermia
 - B. Metabolic acidosis
 - C. Hypokalemia
 - D. Metabolic alkalosis
- ANSWER: B

150. A 23-year-old female patient is scheduled to undergo open reduction and internal fixation of a femoral bone fracture. Six months back, she sustained multiple traumas that resulted in spinal injury at the level of C4 and pelvic injury. Since then, she has been suffering from headache, blurred vision, and nasal stuffiness. During patient assessment, there is an increase in blood pressure and a decrease in heart rate. Laboratory investigation shows normal CBC and urine catecholamine levels. The patient has no known allergic history.

What is the most appropriate anesthesia management plan for this patient?

- A. Rapid sequence induction using an ordinary laryngoscope
- B. Deep level of anesthesia with fiberoptic intubation
- C. Combined epidural anesthesia and spinal anesthesia
- D. General anesthesia under spontaneous ventilation using LMA

ANSWER: B

151. A 42-year-old male patient is brought to an operation room 72 hours after lower extremities and pelvic bone fractures. The surgical team undergoes surgery under general anesthesia with a tourniquet. Immediately after the release of the tourniquet, he developed dyspnea, confusion, petechial rash, hypotension, and arrhythmia. He has no known allergic history. The laboratory findings show a low hemoglobin level and no major abnormality in the patient's serum electrolyte value.

What is the most appropriate diagnosis of this patient?

- A. Fat embolism syndrome
- B. Central pontine myelolysis
- C. Characinoid syndrome
- D. Anaphylactoid reaction

ANSWER: A

152. A 34-year-old female patient is presented to an operation room for open reduction and fixation of femoral shaft fracture secondary to a car accident. She has refused spinal anesthesia due to the recall of the previous noisy intraoperative sound and the pain she had experienced during the previous surgery.

What is the most appropriate strategy to manage this patient's concern?

- A. Postpone the surgery until the patient gives consent for RA
- B. Premedicate with morphine and proceed with RA
- C. Engage in frank discussion and reassurance for alleviating her concern
- D. Use TIVA (Total Intravenous Anesthesia) as alternative management

ANSWER: C

153. A 40-year-old male patient is scheduled to undergo tendon laceration repair of the left hand. Complete anesthesia in a median, ulnar, and radial nerve distribution with axillary block has been achieved. A two-hour inflation of a tourniquet is required to complete the procedure.

What is the most likely nerve to be blocked to cover tourniquet pain for this procedure?

- A. Lateral antebrachial cutaneous nerve
- B. Axillary nerve
- C. Intercostobrachial nerve
- D. Musculocutaneous nerve

ANSWER: C

154. A 46-year-old male patient is brought to an emergency outpatient department two hours after sustaining a car accident injury. The patient has extensive soft tissue injury at his lower extremity. The radiologic finding confirms hip fracture, and she is scheduled for right hip arthroplasty under regional anesthesia. After three days of surgery in the ward, the patient suddenly developed hypoxia, hypotension, and cardiac arrhythmia.

What is the most likely diagnosis of this patient?

- A. Fat embolism
- B. Crush Syndrome
- C. Compartment Syndrome
- D. Bone cement implantation syndrome

ANSWER: D

155. A 54-year-old male patient is brought to an emergency outpatient department four hours after sustaining a car accident injury. The patient has extensive soft tissue injury at his lower extremity. The radiologic finding suggests femoral shaft fracture. After preoperative evaluation, the patient underwent nailing under epidural anesthesia. The patient has been taking a low dose of low molecular weight heparin once per day for DVT prophylaxis.

What is the most appropriate time interval to remove the neural catheter?

- A. 4 to 6 hours after the previous dose
- B. 6 to 8 hours after the previous dose
- C. 8 to 10 hours after the previous dose
- D. 10 to 12 hours after the previous dose

ANSWER: D

156. A 13-year-old male patient presented to an emergency outpatient department two hours after sustaining a thermal burn injury on the anterior torso of the patient. During evaluation, he complains of pain. The vital signs are blood pressure of 150/90 mmHg, heart rate of 135/min, respiratory rate of 25/min, and temperature of 36.5 degrees Celsius.

What is the most appropriate pain management plan for this patient?

- A. Treatment with strong opioid (morphine)
- B. Use of Balanced Analgesia
- C. Use of regional anesthesia
- D. Use of non-steroidal anti-inflammatory drugs

ANSWER: B

157. A senior anesthetist observed that the majority of the women who delivered in a hospital through cesarean section developed surgical site infection. Hence, the anesthetist reviewed literature to check whether there were studies conducted on the problem. However, the anesthetist did not get enough literature on the issue, so the anesthetist decided to conduct research on that.

What is the most likely research title developed by the senior anesthetist in this case?

- A. Proportion of women who delivered by cesarean section in the hospital
- B. Explore the reason for cesarean section delivery among women in the hospital
- C. Factors associated to cesarean section delivery among women in the hospital
- D. Determinants of surgical site infection among women who delivered by cesarean section

ANSWER: D

158. A graduated anesthesia care provider developed a research proposal on the magnitude of back pain and associated factors among mothers who gave birth under spinal anesthesia. After an in-depth analysis of the research problem, he formulated the research objective.

What is the most appropriate specific objective for this study?

- A. To assess the magnitude of back pain and associated factors among mothers who gave birth under spinal anesthesia
- B. To identify factors associated with back pain among mothers who gave birth under spinal anesthesia
- C. To appreciate the knowledge and attitude of mothers who gave birth under spinal anesthesia towards back pain.
- D. To study back pain management practice and associated factors among women who gave birth under spinal anesthesia

ANSWER: B

159. A researcher planned to conduct a study on the association between cigarette smoking and lung cancer among adults in a district.

What is the most likely independent variable for this study?

- A. Lung cancer
- B. Certain districts
- C. Adult population
- D. Cigarette smoking

ANSWER: D

160. A city administration health bureau wishes to find out the magnitude of antiretroviral therapy (ART) treatment adherence and associated factors in the public hospitals. There are eight public hospitals providing ART services in the city administration. From these, three hospitals are randomly selected for the study.

What is the most likely source population for this study?

- A. All HIV patients on ART follow-up in the selected public hospitals
- B. All HIV patients on ART follow-up in selected private and public hospitals
- C. All HIV patients on ART follow-up in public hospitals of the city administration
- D. All HIV patients on ART follow-up in private and public hospitals of the city administration

ANSWER: C

161. An anesthesia department planned to carry out a small-scale study with a relatively short duration because little is known about the problem.

What is the most appropriate epidemiological study design for this study?

- A. Analytical design
- B. Descriptive design
- C. Exploratory design
- D. Interventional design

ANSWER: C

162. A researcher planned to conduct a study about the nutritional status of school adolescent girls. The researcher decided to select specific schools from the total to select a sample of 422 school adolescent girls with an assumption that schools have homogeneous characteristics.

What is the best sampling technique in this case?

- A. Simple random sampling
- B. Cluster random sampling
- C. Stratified random sampling
- D. Systematic random sampling

ANSWER: B

163. As part of clinical services and the quality improvement process, the head of the anesthesia department conducted an assessment on the structure, process, and outcome of care against the agreed explicit standards through a systematic review.
What is the most important benefit of this assessment?
- A. Planning for audit
 - B. Standard selection
 - C. Improving patient care and outcomes
 - D. Formulation and execution of quality strategies
- ANSWER: C
164. A researcher working in a public hospital wants to determine the magnitude of severe acute malnutrition among infants admitted to a pediatric intensive care unit. The researcher proposed to collect data from the patient registration book.
What is the most appropriate data collection technique for this study?
- A. Observation
 - B. Record review
 - C. In-depth interview
 - D. Self-administered questionnaire
- ANSWER: B
165. A researcher plans to conduct a study about community perception towards urban health extension services. The researcher wants to obtain in-depth information on concepts and ideas perceived by different groups in the community.
What is the most appropriate data collection technique to be used by the researcher?
- A. Telephone interview
 - B. Direct observation
 - C. Focus group discussion
 - D. Self-administered questionnaire
- ANSWER: C
166. A researcher studied the effect of folic acid supplementation on pregnancy outcomes using an interventional study design. The researcher respects the participant's capacity for self-determination to participate or not participate in the research.
What is the most likely ethical principle that the researcher applied?
- A. Justice
 - B. Autonomy
 - C. Beneficence
 - D. Non-maleficence
- ANSWER: B
167. A graduating anesthesia student recorded the birth weight of newborns who were delivered in a hospital through cesarean section. The birth weight of newborns in grams was 3200, 2581, 3245, 2069, 2838, 2834, 2759, 2841, 3031, 3101, and 3248.
What is the median weight of the newborns in grams?
- A. 3031
 - B. 2834
 - C. 2841
 - D. 2838
- ANSWER: C

168. A nurse who works in a hospital reported that the mean birth weight of newborns who were delivered in a specific period of time is 3100 grams with a standard deviation of ± 100 grams. What is the most likely interpretation of the reported standard deviation?
- A. An average deviation of each newborn birth weight is 100 grams from the mean birth weight
 - B. An average deviation of each newborn birth weight is 100 grams from the median birth weight
 - C. An average deviation of each newborn birth weight is 100 grams from the middle 50 percent birth weight
 - D. An average deviation of each newborn birth weight is 100 grams between the smallest and highest
- ANSWER: A
169. A researcher carries out a study about the predictors of Myocardial Infarction (MI) among diabetic patients. The researcher conducted a longitudinal follow-up study using 250 exposed and 500 non-exposed study participants. Five years later, the researcher identified 75 new cases from the exposed group and 61 new cases from the non-exposed group. What is the incidence of MI among cases?
- A. 18 percent
 - B. 12 percent
 - C. 30 percent
 - D. 41 percent
- ANSWER: C
170. A department head of anesthesia estimated that the average grade of graduating class anesthesia students is 2.5. Using this assumption, the head took a sample of 36 students from the total and took an assumption of a 95 percent confidence interval and a standard deviation of 6. The interval estimation was reported to be between 1.54 and 3.46. What is the most likely conclusion about the finding?
- A. It is true that the average grade of graduating class students is 2.5
 - B. The average grade of graduating class student is different from 2.5
 - C. It is true the average grade of sampled students was found between 1.54 and 3.46
 - D. The average grade of graduating class student is different from greater than 2.5
- ANSWER: A
171. A student from a graduating class of the anesthesia department is assigned to a small village to assess community health problems. The student observed a significant number of women with clinically evident goiter, for which little attention was given by the women. The student noticed that there is a general hospital in the nearby area that is capable of providing major operation services. What is the most likely health belief model describing the action of the women in this case?
- A. A. High perceived benefits
 - B. B. High Self-Efficacy
 - C. C. Low perceived severity
 - D. D. High perceived susceptibility
- ANSWER: C
172. An anesthetist planned to have detailed discussions with community leaders of 11 sub-cities in a city to explore neglected and undiagnosed operable surgical problems in the locality. What is the most likely need assessment method to be applied in this case?
- A. In-depth interview
 - B. Focus group discussion
 - C. Survey questionnaire
 - D. Delphi Technique
- ANSWER: B

173. A particular association of anesthetists is planning a community-based health promotion program on anesthesia and surgical care. The association wants to prioritize the most important community problem related to anesthesia and surgical care. For a genuine priority-setting of the problems, the association planned to have a stakeholders meeting.

Who is the most likely final decision maker in this regard?

- A. The policy-makers
- B. The community
- C. The health authorities
- D. The health professionals

ANSWER: B

174. An anesthetist wants to segregate wastes generated from a major operation where a three-bin system is used to segregate the waste. The wastes include highly infectious waste, sharps, chemical waste, and general health care waste.

What is the most appropriate segregation method for this case?

- A. Using a brown bin for general waste
- B. Using a yellow bin for infectious waste
- C. Using a black bin for chemical waste
- D. Using a red bin sharps

ANSWER: B

175. A pre-hospital health care worker is asked to design an appropriate preventive strategy for rapidly increasing road traffic accidents in a city. The city is known for having a well-designed road infrastructure and a high prevalence of substance addiction.

What is the most effective preventive strategy for the above scenario?

- A. Helmet utilization
- B. Seatbelt utilization
- C. Blood alcohol level test
- D. Building speed breaker

ANSWER: C

176. A particular city administration is concerned about the increasing prevalence of physical inactivity, use of Ultra-Processed Foods (UPF), cigarette smoking, and harmful use of alcohol among its population.

What is the most likely effect of these risk factors?

- A. Increasing of infectious disease
- B. Increasing of communicable disease
- C. Increasing of non-communicable disease
- D. Increasing of neglected tropical diseases

ANSWER: C

177. A Community Health Worker (CHW) is educating freshman college students on the impact of substance use and its adverse effects on their health and academic performance. The health education involved a personal reflection of a senior student who was substance addicted and dropped out of school as a result of substance use. Currently, the senior student is free from any substance.

What is the most likely health education approach that the CHW applied in this case?

- A. Group discussion
- B. Demonstration
- C. Role modeling
- D. Health Talks

ANSWER: C

178. Repeated outbreaks of malaria are reported from a particular rural village despite the district health office's effort to distribute adequate insecticide-treated bed nets and dry up of stagnant water in the village. The office assigned a public health expert to formulate an effective malaria prevention strategy in this village.

What is the most appropriate measure to be taken by the public health expert?

- A. Residual DDT spray
- B. Environmental manipulation
- C. Anti-malarial chemoprophylaxis
- D. Awareness creation on bed net use

ANSWER: D

179. A 32-year-old male patient presented for open reduction and internal fixation of a right leg femoral fracture. On preoperative assessment, the anesthetist noted that the patient required blood, but the patient refused to take blood because of religious factors. Then, the anesthetist obtained informed consent by agreeing with the patient's decision not to take blood.

What is the most likely ethical principle exercised in this case?

- A. Non-maleficence
- B. Beneficence
- C. Autonomy
- D. Justice

ANSWER: C

180. A 15-year-old male patient presented to an emergency surgery for appendectomy accompanied by his father. In the preoperative informed consent process, the anesthetist disclosed the benefits, risks, and available alternatives related to anesthesia. Even though the patient has understood the information and agreed to the proposed anesthesia, his father refused to consent.

What is the most likely factor influencing the decision of this patient?

- A. Patient's age
- B. Being an emergency
- C. Type of surgery
- D. Level of anxiety

ANSWER: A

181. A 26-year-old mother complains severe headache after having a cesarean section under spinal anesthesia. The anesthetist informed her that the headache may be related to anesthesia and recommended rest, hydration, and anti-pain. However, the patient argued with the anesthetist because she was not informed about the possibility of such a headache during the informed consent process.

What is the most likely malpractice claim in this scenario?

- A. Practice beyond the scope
- B. Breach of confidentiality
- C. Breach of truthfulness
- D. Inadequate disclosure

ANSWER: D

182. A young anesthetist who has been suspected of opioid addiction is working in the post-anesthesia care unit of orthopedic surgery. Patients usually complain of pain despite having strong opioid prescriptions. When the team made a postoperative visit, appropriate analgesics were ordered.

What is the most likely reason for the complaint of this patient?

- A. Poor efficacy of prescribed drugs
- B. Drugs diverted from patients
- C. Incorrect prescription of opioids
- D. Incorrect doses of opioids used

ANSWER: B

183. A female senior anesthetist has been working in a busy specialized hospital for the last 15 years. She was known for her excellent commitment, compassion, and respectful care. Recently, she started to show a lack of motivation, poor empathy, and exhaustion. As a result, the hospital management allowed her a leave for vacation.

What is the most likely threat to the compassionate care of this anesthetist?

- A. Compassion fatigue
- B. Facility context
- C. Self-reported stress
- D. Lack of recognition

ANSWER: A

184. Surgical patients visiting a general hospital are claiming disrespectful care provided by anesthesia professionals. The anesthetists admitted the complaint and claimed that there is a lack of screens for patient privacy, toilets and other resources to provide respectful care to their patients in the hospital.

What is the most likely threat to respectful care in this scenario?

- A. Patient factors
- B. Healthcare setting
- C. Professionals' attitude
- D. Communication gap

ANSWER: B

185. A 70-year-old patient claims the anesthetist was disrespectful during anesthesia care for prostatectomy. The anesthetist was known for the equal respect he shows for all people irrespective of any personal or community characteristics. But the patient explains that the anesthetist did not respect his dignity as he has been respected in the community.

What is the most likely disrespectful care shown by the anesthetist?

- A. Respect for the dignity of human beings
- B. Respect for dignity of personal identity
- C. Respect for the dignity of the patient's religion
- D. Respect for patient's dignity of merit

ANSWER: D

186. An anesthesia team leader in a general hospital has identified a gap in knowledge, skill, and attitude among anesthetists regarding emerging innovations and anesthesia practices. He noticed that if this gap is not bridged, the quality of anesthesia service will be compromised.

What is the most appropriate solution to be applied for this case?

- A. Improve communication skills of anesthetists
- B. Design continuing professional development
- C. Design monitoring and evaluation strategy
- D. Improve interprofessional collaboration

ANSWER: B

187. A 56-year-old male patient, who underwent emergency laparotomy for small bowel obstruction, is suffering from aspiration pneumonia. The morbidity and mortality team calls the anesthetist to explain why the complication occurred. The anesthetist explained that he was alone and the case was an emergency, which needed cricoid pressure application. He requested the surgeon and nursing team to apply cricoid pressure, but all of them refused by explaining that it is not their scope of work.

What is the most likely cause of complication in this patient?

- A. Lack of communication
- B. Lack of collaboration
- C. Lack of competence
- D. Lack of confidence

ANSWER: B

188. A young anesthetist working in a pre-anesthesia clinic is evaluating a 45-year-old male patient presented for varicose vein repair. The anesthetist is asking the patient a relevant history but has no eye contact with the patient. Then, the patient decides not to reply to further inquiries, hoping to cancel his surgery.

What is the most likely component of communication missed in this case?

- A. Nonverbal communication
- B. Listening to patient's views
- C. Involving family members
- D. Verbal communication

ANSWER: A

189. A 48-year-old male patient scheduled for hernia repair visited the pre-anesthesia clinic one day before his surgery. The patient is very angry and emotional because of his lost chart and long waiting time. When the anesthetist asked his name, he said "you can read it from the chart."

What is the most appropriate strategy to evaluate this patient?

- A. Advise him to see another anesthetist
- B. Advise him to change the hospital
- C. Acknowledge his stress
- D. Appoint him for another day

ANSWER: C

190. A 45-year-old female patient, who underwent orthopedic surgery, filed the surgeon and anesthetist with a malpractice claim after two years of surgery. The surgeon and anesthetist have forgotten what happens during the operation as they manage a number of cases daily.

What is the most appropriate way to disprove guilt in this scenario?

- A. Appointing attorney
- B. Consent for surgery
- C. Record keeping
- D. Expert witness

ANSWER: C

191. A human resource manager of a new general hospital and his team are in the phase of recruiting, selecting, appointing, and giving orientation. The management team ensured adequate numbers of skilled healthcare professionals were hired to accomplish the institutional goal.

What is the most likely management function applied in this case?

- A. Organizing
- B. Staffing
- C. Directing
- D. Controlling

ANSWER: B

192. A head of the anesthesia department set a goal to decrease a case cancellation rate from 20 percent to less than 5 percent within the next 6 months. To achieve this, the department head arranged a pre-anesthesia clinic, requested additional anesthetists, and communicated with biomedical departments for maintenance of the necessary equipment.

What principle of management is applied in the above case?

- A. Management by Objectives
- B. Convergence of Work
- C. Division of Labor
- D. Substitution of Resources

ANSWER: A

193. A primary hospital management team, together with the department heads, is preparing a plan for the next 10 years. The team is discussing and writing a statement that describes the hospital's purpose in the community, and the what and why of the hospital's activities.
What is the most likely planning step the management teams are writing in this case?
- A. Vision statement
 - B. Value statement
 - C. Mission statement
 - D. Philosophy statement
- ANSWER: C
194. A team of surgeons and anesthesiologists wants to expand the surgical center in a general hospital. The team was conducting a SWOT analysis and identified that there were adequately trained human resource staff and resources in the hospital.
What is the most likely SWOT component shown in the above case?
- A. Strength
 - B. Weaknesses
 - C. Opportunities
 - D. Threats
- ANSWER: A
195. A newly inaugurated hospital management team planned to attract senior anesthetists from other health institutions. The vacancy was announced through different channels and platforms to make sure prospective employees apply for the job. By the due date of the vacancy, adequate pools of applicants were achieved for the position.
What human resource management principle is most likely applied in this case?
- A. Selection
 - B. Acquisition
 - C. Retention
 - D. Recruitment
- ANSWER: D
196. Anesthetists working in a certain hospital faced difficulty of assessing and examining patients due to a lack of a pre-anesthesia clinic. This caused demotivation among anesthetists in the hospital. The new department head of anesthesia promised to facilitate the opening of a new pre-anesthesia clinic in the hospital. The department head carried out the promise and succeeded in opening a pre-anesthesia clinic within a month.
What is the most likely leadership characteristic possessed by the head?
- A. Commitment
 - B. Courage
 - C. Optimism
 - D. Generosity
- ANSWER: A
197. A department head of anesthesia promotes a shared vision and ownership of outcomes and problems. Important plans and decisions are made with the team. However, the department head allows the more vocal or more outspoken anesthetists to dominate decision-making.
What is the most appropriate leadership styles shown in the above case?
- A. Autocratic
 - B. Paternalistic
 - C. Democratic
 - D. Laissez-faire
- ANSWER: C

198. A department head of anesthesia at a private hospital observed the increment of surgical site infection in a postoperative ward. The department head proposes many optional causes for the condition, namely, breach of sterility technique in the operation room, lack of sterility while dressing the wound, increment of dust and flies in the ward. The department head wanted to assess the risk associated with the condition.

What is the next appropriate decision-making step to be followed in this case?

- A. Diagnose
- B. Set objective or goal
- C. Implementation
- D. Evaluation

ANSWER: A

199. A five-year project that focuses on quality of pain management was launched in a general hospital. The department head of anesthesia is the leader of the project. The project is in the second quarter of the year-2 implementation period. This week, the project lead performed checks on budget utilization, activity performance, and timeline utilization according to the initial plan.

What is the most likely activity performed by the project lead?

- A. Surveillance
- B. Implementation
- C. Monitoring
- D. Evaluation

ANSWER: C

200. A project team working on a surgical site infection control program had given several rounds of training for anesthetists about the identification and reduction of risks associated with surgical site infection. For these activities, the project team conducted an evaluation study to determine the amount of resources used and the number of anesthetists trained.

What most appropriate monitoring and evaluation component the team needs to evaluate?

- A. Output
- B. Process
- C. Outcome
- D. Impact

ANSWER: B